KCSE CHEMISTRY PAPER 2 PAST PAPERS 2006-2019

K.C.S.E 2006 CHEM PP2 QUESTIONS

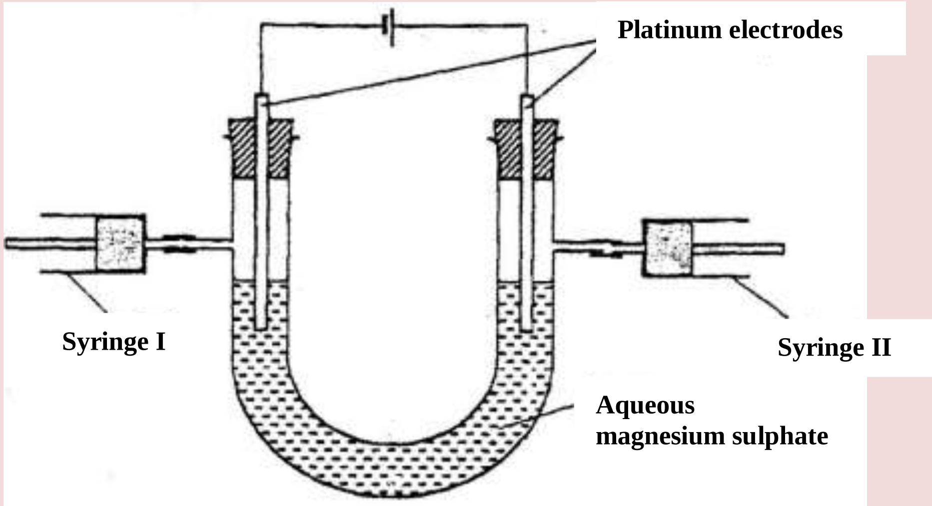
1. a) What is an electrolyte? (1mk)

   b) State how the following substances conduct electricity.

      i) Molten calcium chloride (1mk)

      ii) Graphite. (1mks)

   c) The diagram below shows a set up that was used to electrolyse aqueous magnesium sulphate.



      i) On the diagram above, using an arrow, show the direction of flow of electrons. (1mk)

      ii) Identify the syringe in which hydrogen gas would be collected. Explain (1mk)

   d) Explain why the concentration of magnesium sulphate was found to have increased at the end of the experiment. (2mks)

   e) During the electrolysis, a current of 0.72A was passed through the electrolyte for 15 munites.Calculate the volume of gas produced at the anode. (1 Faraday = 96 500 coulombs; molar gas volume is 24000cm^3 at room temperature). (4mks)

2. a) In an experiment to determine the molar heat of reaction when magnesium displaces copper ,0.15g of magnesium powder were added to 25.0cm^3 of 2.0M copper (II) chloride solution. The temperature of copper (II) chloride solution was 25^{o}C.While that of the mixture was 43^{o}C.

   i) Other than increase in temperature, state and explain the observations which were made during the reaction. (3mks)

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ii) Calculate the heat change during the reaction (specific heat capacity of the solution = $4.2jg^{-1}k^{-1}$and the density of the solution = $1g/cm^3$ (2mks)

iii) Determine the molar heat of displacement of copper by magnesium.(Mg=24.0).

iv) Write the ionic equation for the reaction. (1mk)

v) Sketch an energy level diagram for the reaction. (2mks)

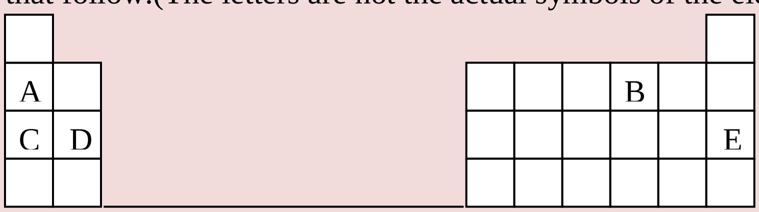
b) Use the reduction potentials given below to explain why a solution containing copper ions should not be stored in a container made of zinc.

$Zn^{2+}_{(aq)} + 2e \longrightarrow Zn_{(s)}$; $E^{\emptyset} = -0.76v$

$Cu^{2+}_{(aq)} + 2e \longrightarrow Cu_{(s)}$; $E^{\emptyset} = +0.34v$ (2mks)

3. a) Distinguish between isotopes and allotropes. (2mks)

b) The chart below is part of the periodic table. Study it and answer the questions that follow.(The letters are not the actual symbols of the elements).



i) Select the element in period three which has the shortest atomic radius. Give a reason for you answer. (2mks)

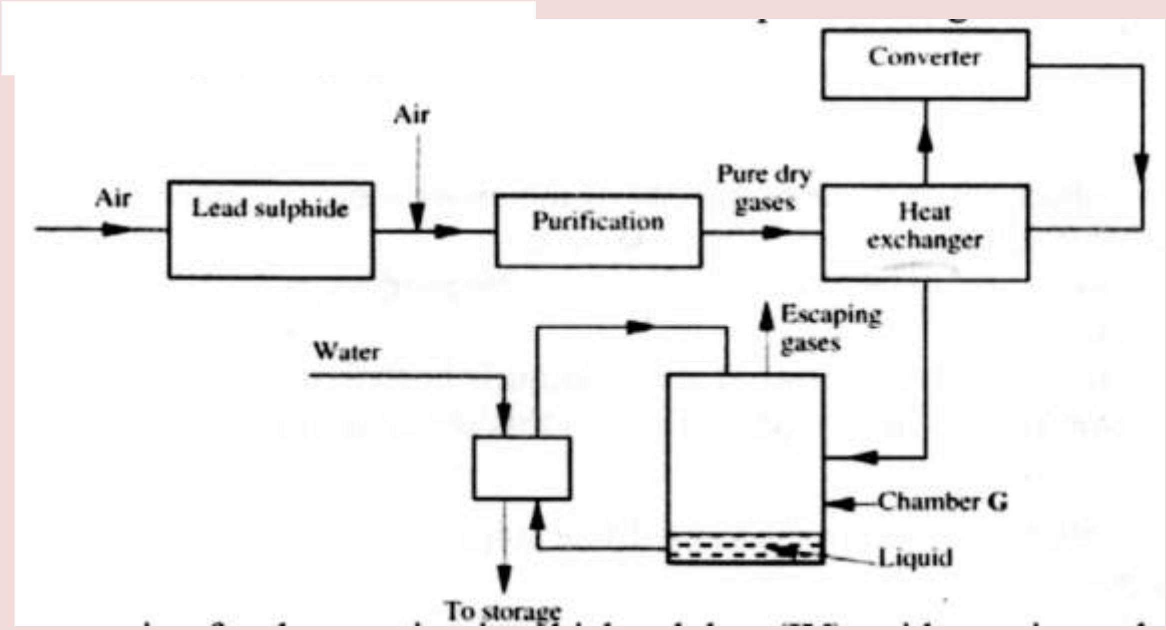
ii) Element F has the electronic structure, 2.8.18.4 on the chart above, indicate the position of element F. (1mks)

iii) State one use of the elements of which E is a member. (1mk)

iv) Write an equation to show the action of heat on the nitrate of element C. (1mks)

c) When 3 litres of chlorine gas were completely reacted with element D, 11.875g of the product were formed.Dertermine the relative atomic mass of element D. (Atomic mass of chlorine = 35.5; molar gas volume = 24litres). (3mks)

4. a) The diagram below shows some processes that take place during the industrial manufacture of sulphuric acid.



(i) Write the equation for the reaction in which sulphur dioxide gas is produced. (1mk)

(ii) Why is it necessary to keep the gases pure and dry ? (1mk)

(iii) Describe the process that takes place in chamber G. (1mk)

(iv) Name the gases that escape into the environment. (1mk)

(v) State and explain the harmful effect on the environment of one of the gases named in (iv) above (1mk)

(vi) Give one reason why it is necessary to use a pressure of 2-3 atmoshperes and not more. (1mk)

b) (i) Complete the table below to show the observations made when concentrated sulphuric acid is added to the substances shown. (2mks)

Substance	Observation
Iron fillings	
Crystals of white sugar	

(ii) Give reasons for the observations made using:

I iron fillings (1mk)

II Crystals of white sugar. (1mk)

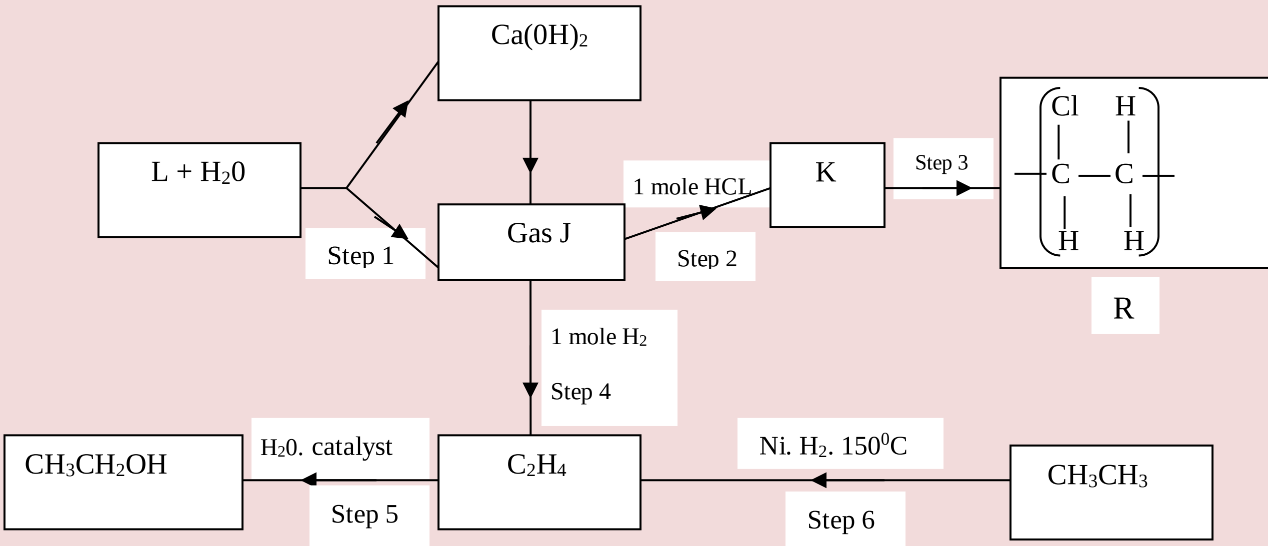
(c) Name one fertilizer made from sulphuric acid. (1mk)

(d) Suggest a reason why $BaSO_4$(apigment made from sulphuric acid) would be suitable in making paint for cars. (1mk)

5. a) What name is given to a compound that contains carbon and hydrogen only? (½mk)

b) Hexane is a compound containing carbon and hydrogen.
(i) What method is used to obtain hexane from crude oil? (1mk)
(ii) State one use of hexane (1mk)

c) Study the flow chart below and answer the questions that follow.



(i) Identify reagent L. (1mk)

(ii) Name the catalyst used in step 5. (1mk)

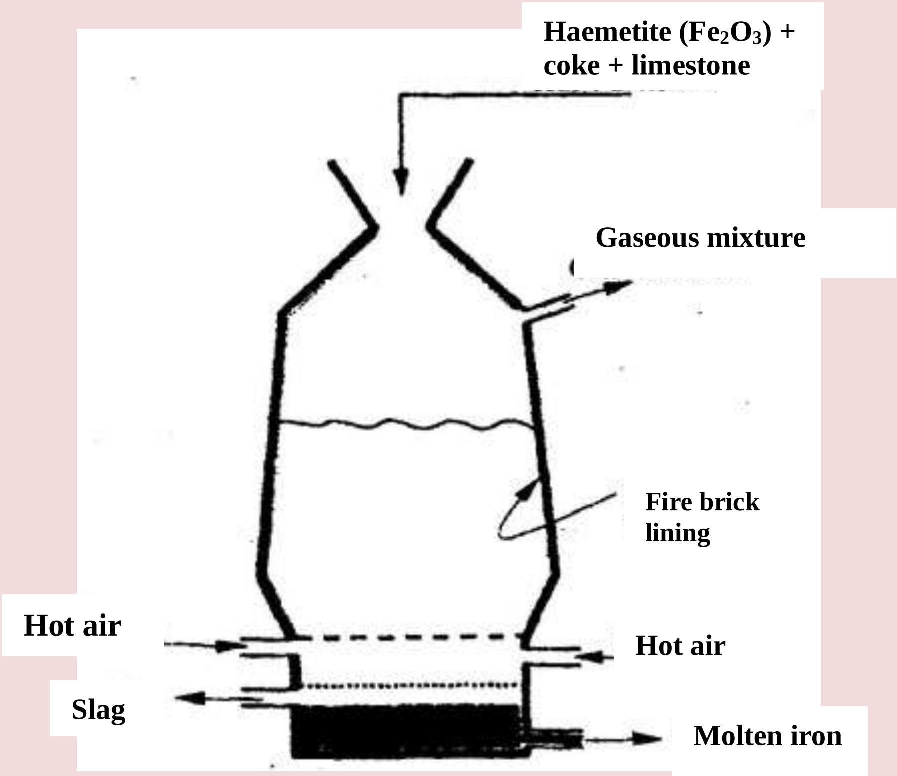
(iii) Draw the structural formula of gas J. (1mk)

(iv) What name is given to the process that takes place in step 5? (½mk)

d) (i) write the equation for the reaction between aqueous sodium hydroxide and aqueous ethanoic acid. (1mk)

(ii) Explain why the reaction between 1g of sodium carbonate and 2M hydrochloric acid is faster than the reaction between 1g of sodium carbonate and 2M ethanoic acid. (1mks)

6. The extraction of iron from its ores takes place in the blast furnace. Study it and answer the questions that follow.



a) Name
   (i) One of the substances in the slag (1mk)
   (ii) Another iron ore material used in the blast furnance. (1mk)
   (iii) One gas which is recycled. (1mk)

b) Describe the process which leacd to the formation of iron in the blast furnace

c) State the purpose of limestone in the blast furnace. (3mks)

d) Give a reason why the melting point of the iron obtained from the blast furnace is 1200^0 C while tat of pure iron is 1535^0C (1mk)

(e) State two uses of steel (2mks)

7. The table below shows the volumes of nitrogen dioxide gas produced when different volume of IM nitric acid were each reacted with 2.07 g of lead at room temperature.

Volume of 1 M nitric acid (cm^3)	Volume of nitrogen dioxide gas (cm^3)
5	60
15	180
25	300
35	420
45	480
55	480

a) Give a reason why nitric acid is not used to prepare hydrogen gas. (1mk)

b) Explain how the rate of the reaction between lead and nitric acid would be affected if the temperature of the reaction mixture was raised. (2mks)

c) On the grid provided below, plot a graph of the volume of the gas produced (Vertical axis) against volume of acid. (3mks)

d) Using the graph, determine the volume of:

i) Nitrogen dioxide produced when 30cm^3 of 1 M nitric acid were reacted with 2.07 g of lead (1mk)

ii) 1M nitric acid which would react completely with 2.07g of lead. (1mk)

e) Using the answer in d(i) above, determine:

i) The volume of 1M nitric acid that would react completely with one mole of lead (pb=207) (2mks)

ii) The volume of nitrogen dioxide gas produced when one mole of lead reacts with excess 1 M nitric room temperature. (1mk)

f) Calculate the number of moles of:

i) 1M nitric acid that reacted with one mole of lead (1mk)

ii) nitrogen dioxide produced when one mole of lead were reacted with excess nitric acid. (Molar gas volume of 2400cm^3) (1mk)

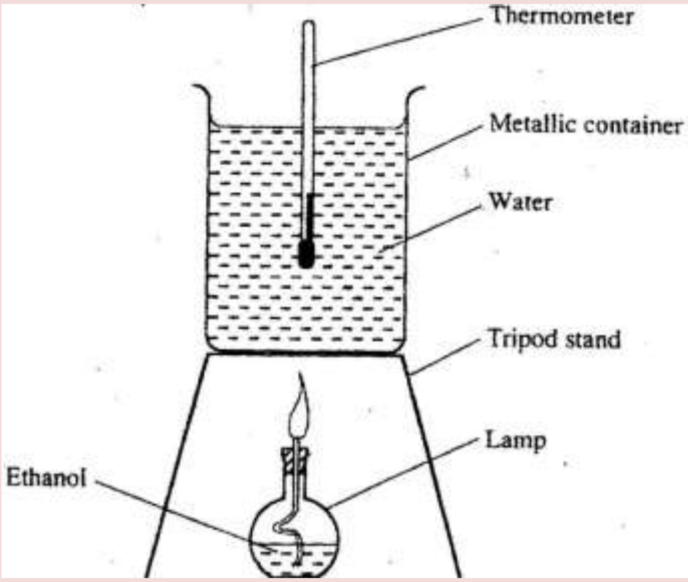
g) Using the answers obtained in f (i) and (ii) above, write the equation for the reaction between lead and nitric acid given that one mole of lead nitrate and two moles of water were also produced. (1mk)

CHEMISTRY PAPER 233/2

K.C.S.E 2007 QUESTIONS

1. (a) State two factors that should be considered when choosing fuel for cooking (2mks)

(b) The diagram below represents a set – up that was used to determine the molar heat of combustion of ethanol



During the experiment, the data given below was recorded

Volume of water	$450cm^3$
Initial temperature of water	25^0 C
Final temperature of water	46.5^0C
Mass of ethanol + Lamp before burning	125.5g
Mass of ethanol + lamp after burning	124.0g

Calculate the:

(i) Heat evolved during the experiment (density of water = $1g/cm^3$
Specific heat capacity of water = 4.2 $Jg^{-1}K^{-1}$ (3mks)

(ii) Molar heat of combustion of ethanol (C = 12.0, O = 16.0, H=1.0) (2mks)

(c) Write the equation for the complete combustion of ethanol (1mk)

(d) The value of the molar heat of combustion of ethanol obtained in (b) (ii) above is lower than the theoretical value. State two sources of error in the experiment. (2mks)

2. (a) Give the systematic names of the following compounds

(i) $CH_2 = C - CH_3$ with CH_3 bonded below C

$$\begin{array}{l} CH_2 = \underset{\displaystyle CH_3}{\underset{|}{C}} - CH_3 \end{array}$$

(1 mk)

(ii) $CH_3CH_2CH_2C \equiv CH$ (1mk)

(b) State the observations made when Propan – I- ol reacts with:

(i) Acidified potassium dichromate (VI) Solution (1mk)

(ii) Sodium metal (1mk)

(c) Ethanol obtained from glucose can be converted to ethane as shown below

$$C_6H_{12}O_6 \xrightarrow{\textbf{Step 1}} C_2H_5OH \xrightarrow{\textbf{Step II}} CH_2 \equiv CH_2$$

Name and describe the process that take place in steps I and II

Step I (1½ mks)

Step II (1½ mks)

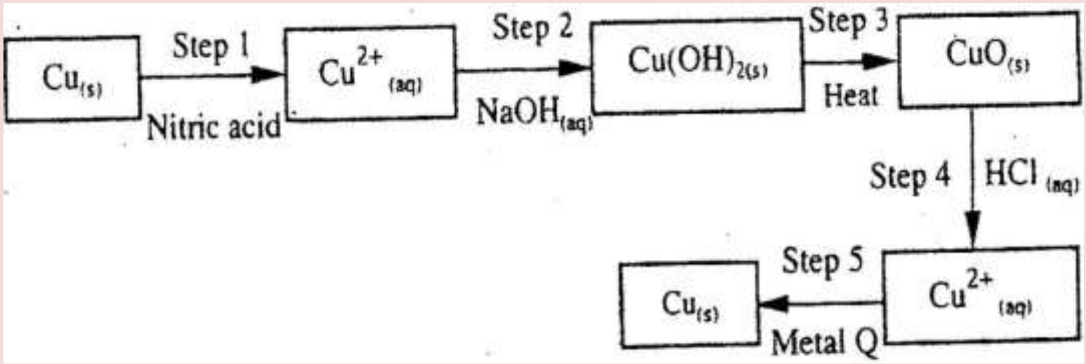
(d) Compounds A and B have the same molecular formula $C_3H_6O_2$. Compound A liberates carbon (IV) oxide on addition of aqueous sodium carbonate while compound B does not. Compound B has a sweet smell. Draw the possible structures of:

(i) Compound A (1 mk)

(ii) Compound B (1 mk)

(e) Give two reasons why the disposal of polymers such as polychloroethane by burning pollutes the environment. (2mks)

3. The flow chart below shows a sequence of chemical reactions starting with copper study it and answer the questions that follow.



(a) In step 1, excess 3M nitric acid was added to 0.5g of copper powder

(i) State two observations which were made when the reactions was in progress (2mks)

(ii) Explain why dilute hydrochloric acid cannot be used in step 1 (1mk)

(iii) I Write the equation for the reaction that took place in step 1 (1mk)

II Calculate the volume of 3M nitric that was needed to react completely with 0.5g of copper powder. (Cu = 63.5) (3 mk)

(b) Give the names of the types of reactions that took place in steps 4 and 5 (1 mk)
Step 4
Step 5

(c) Apart from the good conductivity of electricity, state two other properties that make it possible for copper to be extensively used in the electrical industry. (2mks)

4. (a) Methanol is manufactured from carbon (IV) oxide and hydrogen gas according to the equation:

$$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g)$$

The reaction is carried out in the presence of a chromium catalyst at 700K and 30kPa. Under these conditions, equilibrium is reacted when 2% of the carbon (IV) oxide is converted to methanol

(i) How does the rate of the forward reaction compare with that of the reverse reaction when 2% of the carbon (IV) oxide is converted to methanol? (1 mk)

(ii) Explain how each of the following would affect the yield of methanol:

I Reduction (2mks)

II Using a more efficient catalyst (2mks)

(iii) If the reaction is carried out at 500K and 30kPa, the percentage of carbon (IV) oxide converted to methanol is higher than 2%

I what is the sign of ΔH for the reaction? Give a reason (2mks)

II Explain why in practice the reaction is carried out at 700K but NOT at 500K (1mk)

(b) Hydrogen peroxide decomposes according to the following equation:

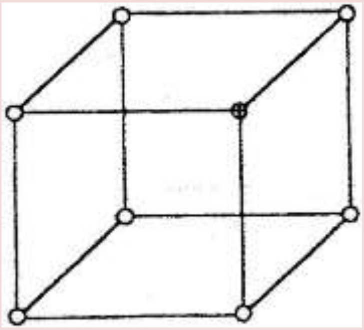
$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2 (g)$

In an experiment, the rate of decomposition of hydrogen peroxide was found to be 6.0×10^{-8} mol dm^{-3} S^{-1}.

(i) Calculate the number of moles per dm^3 of hydrogen peroxide that had decomposed within the first 2 minutes (2mks)

(ii) In another experiment, the rate of decomposition was found to be 1.8×10^{-7} mol $dm^{-3}S^{-1}$. The difference in two rates could have been caused by addition of a catalyst. State, giving reasons, one other factor that may have caused the difference in two rates of decomposition (2mks)

5. (a) The diagram below represents part of the structure of a sodium chloride crystal. The position of one of the sodium ions in the crystal is shown as ⊕



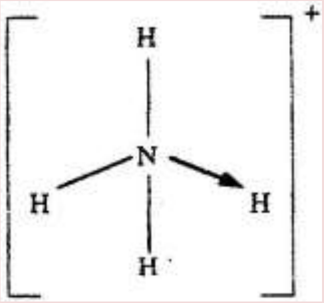
(i) On the diagram, mk the position of the other three sodium ions (2mks)

(ii) The melting and boiling points of sodium chloride are 801^0C and 1413^0C respectively.

Explain why sodium chloride does not conduct electricity at 25^0C, but does so at temperatures between 801^0 C and 1413^0C (2mks)

(b) Give a reason why ammonia gas is highly soluble in water (2mks)

(c) The structure of an ammonia ion is shown below:



Name the type of bond represented in the diagram by N → H (1 mk)

(d) Carbon exists in different crystalline forms. Some of these forms were recently discovered in soot and are called fullerenes

(i) What name is given to different crystalline forms of the same element? (1mk)

(ii) Fullerenes dissolve in methylbenzene while the other forms of carbon do not.
Given that soot is a mixture of fullerenes and other solid forms of carbon, describe how crystals of fullerenes can be obtained from soot. (3mks)

(iii) The relative molecular mass of one of the fullerenes is 720. What is the molecular formula of this fullerene? (C=12.00 (1 mk)

6. (a) The elements nitrogen, phosphorous and potassium are essential for plant growth.

(i) Potassium in fertilizers may be in the form of potassium nitrate
Describe how a sample of a fertilizer may be tested to find out if it contained nitrate ions. (2mks)

(ii) Calculate the mass of nitrogen present if a 25kg bag contained pure ammonium phosphate, $(NH_4)_2 HPO_4$.
(N = 14.0, H=1.0, P = 31.0, O = 16.0 (2mks)

(b) The table below shows the solubility of ammonium phosphate in water at different temperatures.

Temperature (C^0)	Solubility of ammonium phosphate in g/100g water
10	63.0
20	69.0
30	75.0
40	82.0
50	89.0
60	97.0

(i) On the grid provided, draw the solubility curve of ammonium phosphate (Temperature on x – axis) (3mks)

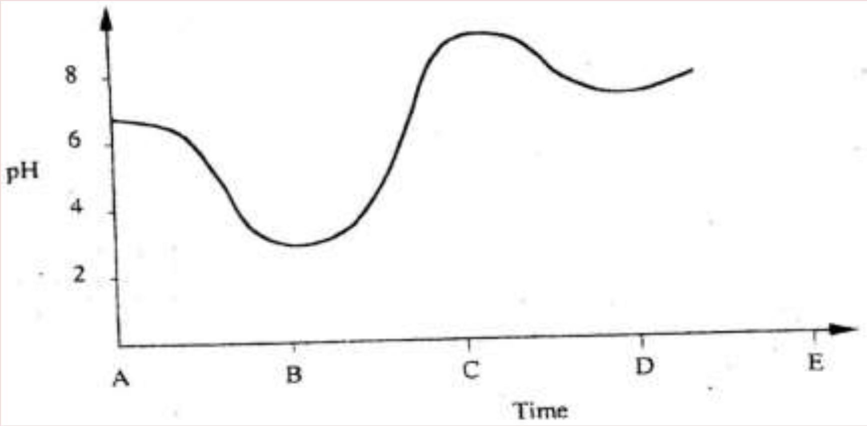
(ii) Using the graph, determine the solubility of ammonium phosphate at 25^0C (1 mk)

(iii) 100g of a saturated solution of ammonium phosphate was prepared at 25^0C

I what is meant by a saturated solution? (1mk)

II Calculate the mass of ammonium phosphate which was used to prepare the saturated solution (2mks)

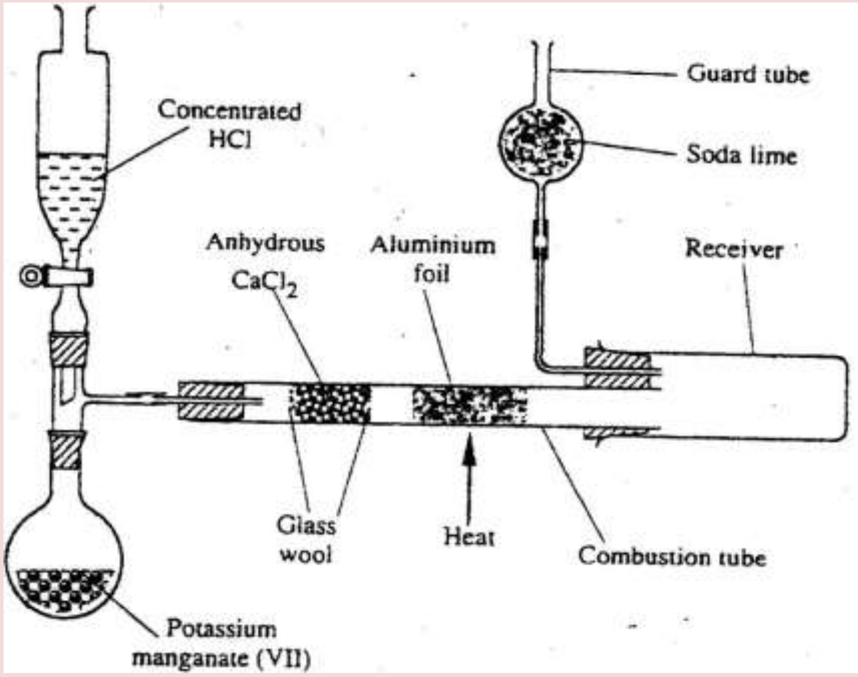
(c) The graph below shows how the PH value of soil in a farm changed over a period of time



(i) Describe how the pH of the soil can be determined (2mks)

(ii) State one factor that may have been responsible for the change in the soil pH in the time interval AB (1 mk)

7. The diagram below shows the set up used in an experiment to prepare chlorine gas and react it with aluminium foil. Study it and answer the question that follow



(a) In the experiment, concentrated hydrochloric acid and potassium manganate (VII) were used to prepare chlorine gas. State two precautions that should be taken in carrying out this experiment. (2mks)

(b) Write the formula of another compound that could be used instead of potassium manganate (VII) (1 mk)

(c) Explain why it is necessary to allow the acid to drip slowly onto potassium manganate (VII) before the aluminium foil is heated. (2mks)

(d) State the property of the product formed in the combustion tube that makes it possible for it to be collected in the receiver (1mk)

(e) When 1.08g of aluminum foil were heated in a stream of chlorine gas, the mass of the product formed was 3.47 g

Calculate the:

(i) Maximum mass of the product formed if chlorine was in excess;

(Al= 27; Cl = 35.5)

(ii) Percentage yield of the product formed (1 mk)

(f) Phosphorous trichloride is a liquid at room temperature. What modification should be made to set up if it is to be used to prepare phosphorous trichloride? (1 mk)

CHEMISTRY PAPER 233/2

K.C.S.E 2008 QUESTIONS

1. a) Biogas is a mixture of mainly carbon (IV) oxide and methane.

(i) Give a reason why biogas can be used as a fuel. (1mk)

(ii) Other than fractional distillation, describe a method that can be used to determine the percentage of methane in biogas. (3mks)

b) A sample of biogas contains 35.2% by mass of methane. A biogas cylinder contains 5.0 kg of the gas.

Calculate the;

(i) Number of moles of methane in the cylinder. (Molar mass of methane=16) (2mks)

(ii) Total volume of carbon (IV) oxide produced by the combustion of methane in the cylinder (Molar gas Volume=24.0 dm^{-3+} at room temperature and pressure). (2mks)

c) Carbon (Iv) oxide, methane, nitrogen (I) oxide and trichlorofluoromethane are green-house gases.

(i) State one effect of an increased level of these gases to the environment. (1mk)

(ii) Give one source from which each of the following gases is released to the environment;

I Nitrogen (i) oxide (1 mk)

II Trichlorofluoromethane. (1mk)

2 a) Write an equation to show the effect of heat on the nitrate of:

(i) Potassium (1mk)

b) The table below gives information about elements $A_1A_2A_{3,}$ and A_4

Element	Atomic Number	Atomic Radius (nm)	Ionic radius (nm)
A_1	3	0.134	0.074
A_2	5	0.090	0.012

A_3	13	0.143	0.050
A_4	17	0.099	0.181

(i) In which period of the periodic table is element A_2? Give a reason (2mks)

(ii) Explain why the atomic radius of:

I A_1 is greater than that of A_2;

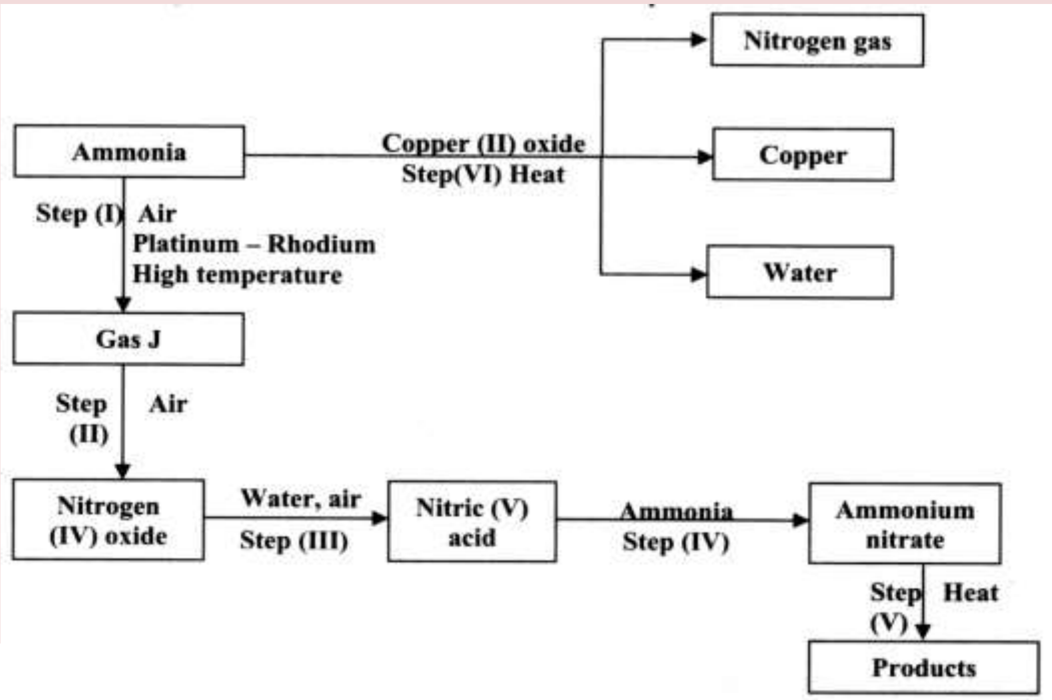
II A_4 is smaller than its ionic radius (2mks)

(iii) Select the element which s in the same group as A_3 (1 mk)

(iv) Using dots (.) and crosses(x) to represent outermost electrons. Draw a diagram to show the bonding in the compound formed when A_1 reacts with A_4 (1 mk)

3. (a) Describe the process by which Nitrogen is obtained from air on a large scale. (4mks)

(b) Study the flow chart below and answer the questions that follow.



(i) Identify gas J. (1 mk)

(ii) Using oxidation numbers, show that ammonia is the reducing agent in step (VI) (2mks)

(iii) Write the equation for the reaction that occurs in step (V). (1mk)

(iv) Give one use of ammonium nitrate. (1mk)

c) The table below shows the observations made when aqueous ammonia was added to cations of elements F_2F and G until in excess.

Cation of	Addition of a few drops of Aqueous ammonia.	Addition of excess aqueous ammonia.
E	White precipitate	Insoluble
F	No precipitate	No precipitate
G	White precipitate	Dissolves

(i) Select the cation that is likely to be Zn^{2+} (1mk)

(ii) Given that the formula of the cation of element E is E^{2+}, write the ionic equation for the reaction between $E^{2+}_{(aq)}$ and aqueous ammonia. (1mk)

4.a) (i) State the Le chatelier's principle. (1mk)

(ii) Carbon (II) oxide gas reacts with steam according to the equation;

$CO_{(g)} + H_2O_{(g)} \longrightarrow H_{2(g)} + CO{+}2_{(g)}$

What would be the effect of increasing the pressure of the system at equilibrium? Explain. (2mks)

b) The table below gives the volumes of oxygen gas produced at different times when hydrogen peroxide decomposed in the presence of a catalyst.

Time(Sec)	0	10	20	30	40	50	60
Volume of oxygen (cm^3)	0	66	98	110	119	120	120

(i) Name the catalyst used for this reaction (1mk)

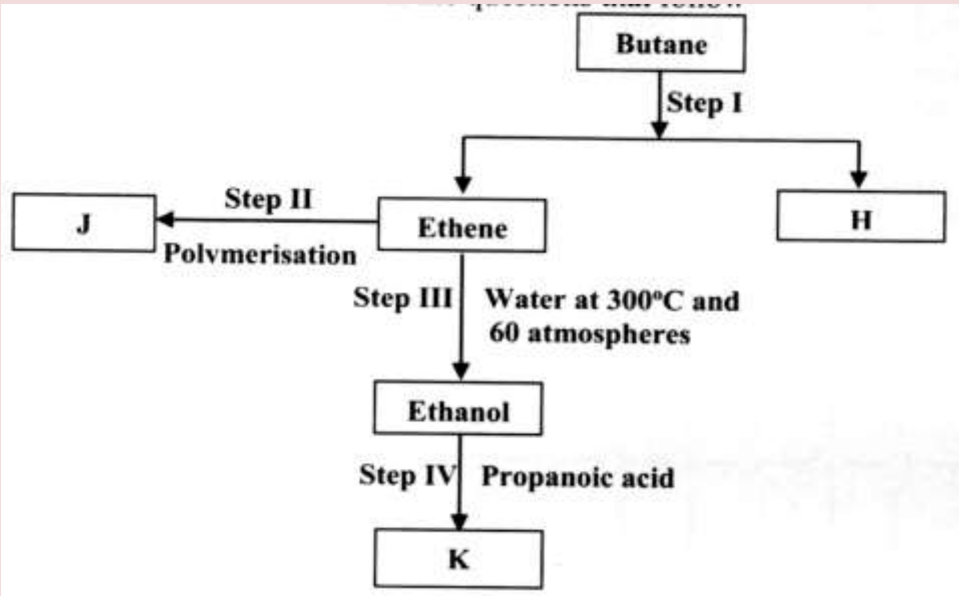
(ii) On the grid provided, draw the graph of volume of oxygen gas produced (vertical axis) against time. (3mks)

(iii) Using the graph, determine the rate of decomposition of hydrogen peroxide after 24 seconds. (2mks)

(iv) Give a reason why the total volume of oxygen gas produced after 50 seconds remains constant. (1mk)

5. (a) Alkanes, alkenes and alkynes can be obtained from crude oil. Draw the structure of the second member of the alkyne homologous series. (1mk)

(b) Study the flow chart below and answer the questions that follow



(i) State the conditions for the reaction in step 1 to occur (1 mk)

(ii) Identify substance II (1 mk)

(iii) Give:

I. One advantage of the continued use of substance such as J (1 mk)

II The name of the process that takes place in step III (1 mk)

III The name and the formula of substance K (2mks)

Name:……………………………

Formula:…………………………

(iv) The relative molecular mass of J is 16,800. Calculate the number of monomers that make up J.

(c) The table below give the formula of four compounds L,M,N and P

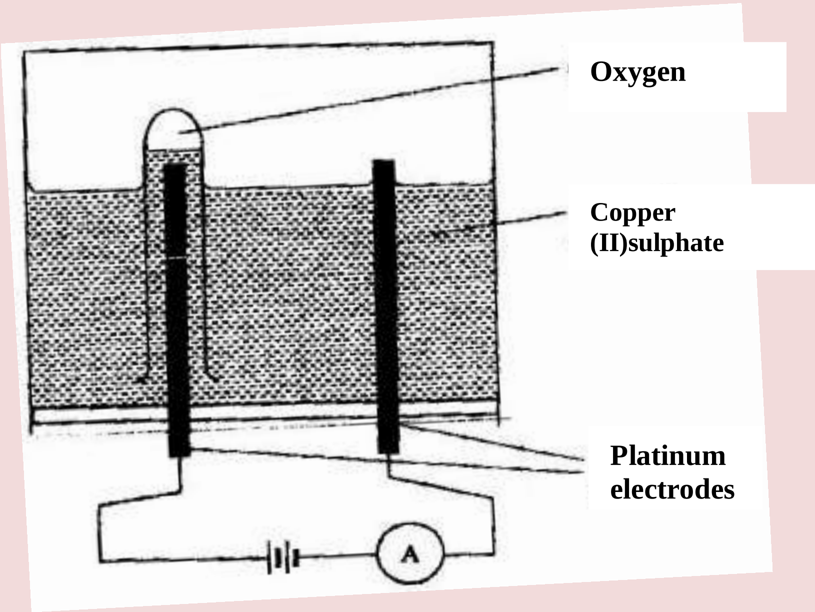
Compound	Formula
L	C_2H_6O
M	C_3H_6
N	$C_3H_6O_2$
P	C_3H_8

Giving a reason in each case, select the letter which represents a compound that:

(i) Decolorizes bromine in the absence of UV light (2mks)

(ii) Gives effervescence when reacted with aqueous sodium carbonate (2mks)

6. The diagram below represents a set up that can be used to electrolyze aqueous copper (II) sulphate.



(a) (i) Describe how oxygen gas is produced during the electrolysis (2mks)

(ii) Explain why copper electrodes are not suitable for this electrolysis (2mks)

(b) Impure copper is purified by an electrolytic process

(i) Name one ore from which copper is obtained (1mk)

(ii) Write the equation for the reaction that occur at the cathode during the purification of copper (1mk)

(iii) In an experiment to electroplate a copper spoon with silver, a current of

0.5 A was passed for 18 minutes. Calculate the amount of silver deposited on the spoon (π = 96500 coulombs, Ag = 108) (3mks)

(iv) Give two reasons why some metals are electroplated (2mks)

7. (a) Define the standard enthalpy of formation of a substance (1mk)

(b) Use the thermo chemical equations below to answer the questions that follow.

1. $C_2H_6(g) + {}^7/_2O_2(g) \rightarrow 2CO_2(g) + 3H_2O\ (i)$; $\Delta\ H_1 – 1560kjmol^{-1}$
2. $C_{(graphite)} + O_2\ (g) \rightarrow CO_2(g)$; $\Delta H_2 = -394kJmol^{-1}$
3. $H_2(g) + ½\ O_2\ (g) \rightarrow H_2O(g)$ $\Delta H_3 = -286kJmol^{-1}$

(i) Name two types of heat changes represented by ΔH_3 (2mks)

(ii) Draw an energy level diagram for the reaction represented by equation 1

(iii) Calculate the standard enthalpy of formation of ethane (2mks)

(iv) When a sample of ethane was burnt, the heat produced raised the temperature of 500g of water by 21. 5 K, (specific heat capacity of water = $4.2Jg^{-1}K$). Calculate the:

I. Heat change for the reaction (2mks)

II. Mass of ethane was burnt. (relative formula mass of ethane= 30) (2mks)

CHEMISTRY PAPER 233/2

K.C.S.E 2009 QUESTIONS

1. a) Two reagents that can be used to prepare chlorine gas are manganese (IV) oxide and concentrated hydrochloric acid.

i) Write an equation for the reaction. (1 mk)

ii) Give the formula of another reagent that can be reacted with concentrated hydrochloric acid to produce chlorine gas. (1 mk)

iii) Describe how the chlorine gas could be dried in the laboratory (2mks)

b) In an experiment, dry chlorine gas was reacted with aluminium as shown in figure 1.

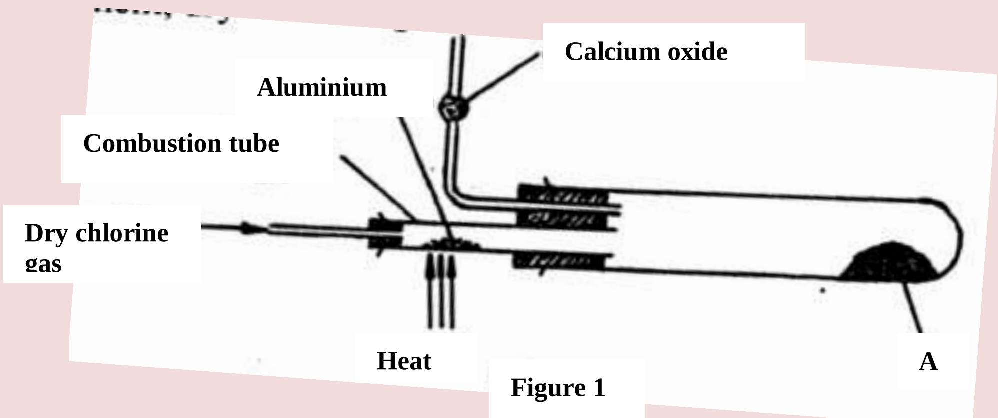


Figure 1

i) Name substance A. (1 mk)

ii) Write an equation for the reaction that took place in the combustion tube. (1mk)

iii) 0.84 g of aluminium reacted completely with chlorine gas. Calculate the volume of chlorine gas used (Molar gas volume is $24dm^3$, al = 27). (3mks)

iv) Give two reasons why calcium oxide is used in the set up. (2mks)

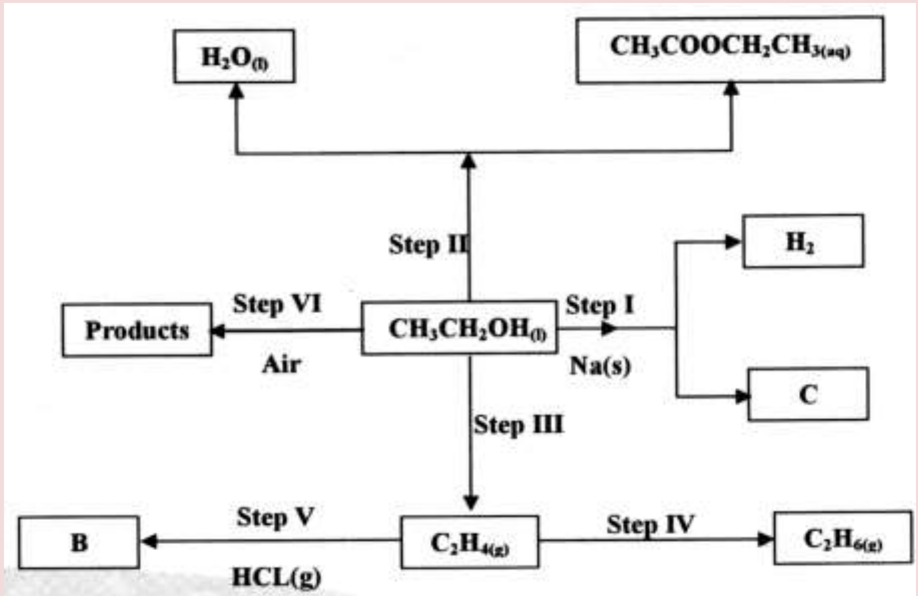
2 a) Draw the structures of the following compounds: (2mks)

i) 2- methylbut -2 ene;

ii) heptanoic acid

b) Describe a physical test that can be used to distinguish between methanol and hexanol. (2mks)

c) Use the flow chart below to answer the questions that follow.



i) Name:

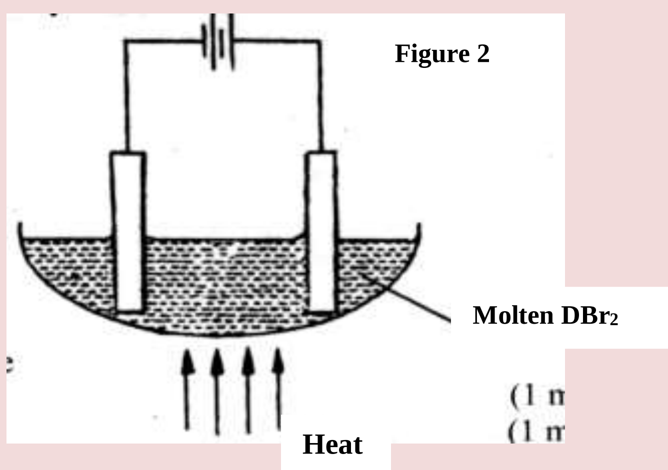
(I) the type of reaction that occurs in step II; (1 mk)

(II) Substance B. (1 mk)

ii) Give the formula of substance C. (1 mk)

iii) Give the reagent and the conditions necessary for the reaction in step (IV) (3mks)

3 The set-up below (figure 2) was used to electrolyse a bromide of metal D DBr_2.



i) Write equation for the reactions at the

I cathode (1 mk)

II anode (1mk)

ii) The electrodes used in the experiment were made of carbon and metal D. which of the two electrodes was used as the anode? Give a reason. (2 mks)

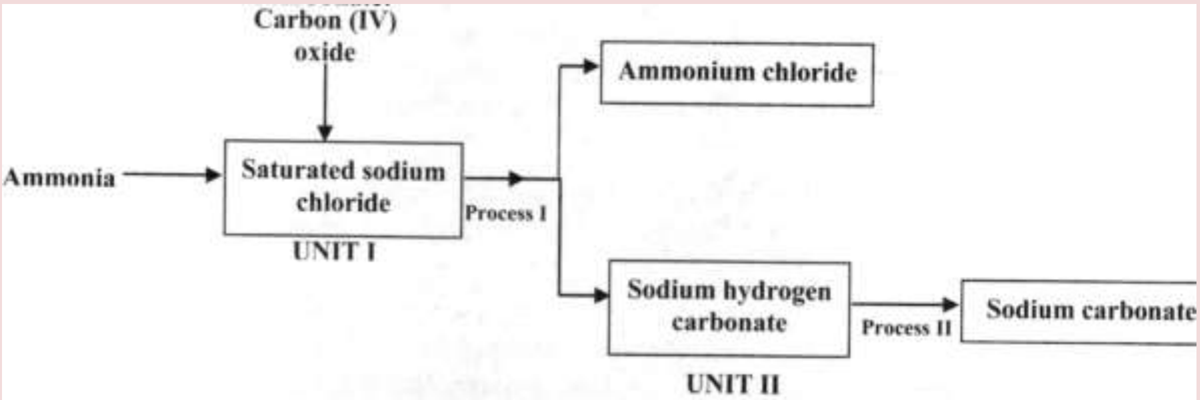
iii) Give a reason why this experiment is carried out in a fume cupboard. (1 mk)

iv) When a current of 0.4 A was passed for 90 minutes, 2.3 1 g of metal D were deposited.

I Describe how the amount of metal D deposited was determines. (3mks)

II Calculate the relative atomic mass of metal D. (I Faraday = 96500 coulombs) (3 mks)

4. a) the schematic diagram shows part of the Solvay process used for the manufacture of sodium carbonate.



i) Explain how the sodium chloride required for this process is obtained from sea water. (2mks)

ii) Two main reactions take place in UNIT I . The first one is the formation of ammonium hydrogen carbonate.

I. Write an equation for this reaction (1 mk)

II. Write an equation for the second reaction (1 mk)

iii) State how the following are carried out: (2mks)

I Process I

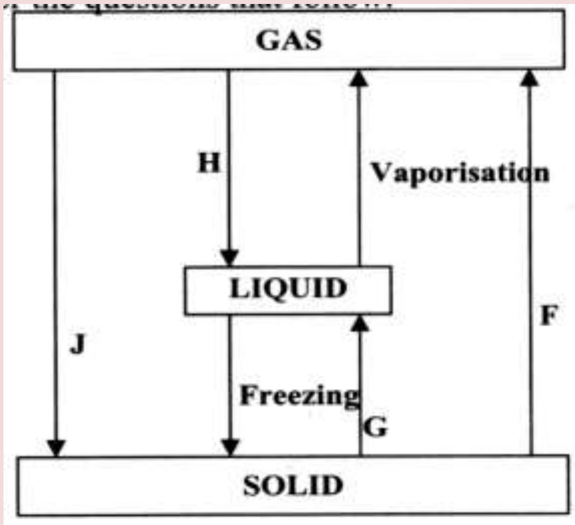
II) Process II

iv) In an experiment to determine the percentage purity of the sample of sodium carbonate produced in the Solvay process, 2.15 g of the sample reacted completely with 40.0cm^3 of 0.5 M sulphuric acid.

I calculate the number of moles of sodium carbonate that reacted. (2mks)

II Determine the percentage of socium carbonate in the sample.
(Na= 23.0, C= 12.0, O = 16.0) (2mks)

b) Name two industrial uses of sodium carbonate (2mks)
have been identified and others labelled.

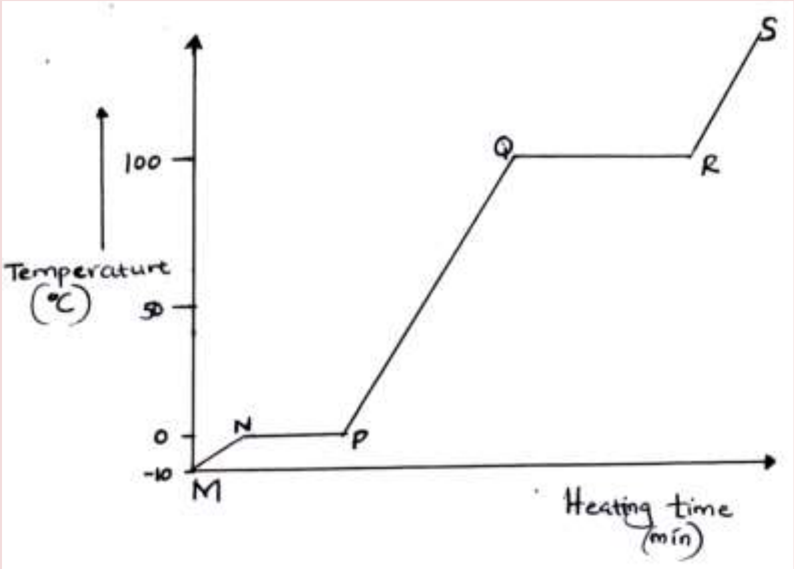


i) Give the names of the processes
I H (1 mk)
II G (1 mk)

ii) Name one substance that can undergo process F when left in an open container in the laboratory. (1 mk)

iii) The process J is called deposition. Using water as an example, write an equation that represents the process of deposition. (1 mk)

b) Figure 4 shows the beating curve for water.



i) Give the names of the intermolecular forces of attraction in the segments;

I MN (1mk)

II RS (1 mk)

ii) The heats of fusion and vaporization of water are 334.4 Jg^{-1} and 1159.4 Jg^{-1} respectively.

I Explain why there is a big difference between the two. (2mks)

II How is the difference reflected in the curve? (1 mk)

c) Coal, oil and natural gas are major sources of energy. They are known as fossil fuels. Hydrogen is also a source of energy.

i) State and explain two reasons why hydrogen is a very attractive fuel compared to fossils. (3mks)

ii) State one disadvantage of using hydrogen fuel instead of fossil fuels.

6. a) Study the table below and complete it. (W^{-1} and X^{4+} are not the actual symbols of the ions). (2mks)

Iron	Number of protons	Number of neutrons	Mass Number	Electron arrangement
W	……………..	20	…………….	2.8.8
X^{4+}	14	……………	28	…………..

b) State the observation that would be made in the following tests to distinguish between:

i) Sodium and copper burning pieces of each in air. (2 mks)

ii) Sodium and Magnesium by placing small pieces of each in cold water which contains two drops of phenolphalein. (2mks)

c) The atomic numbers of Na and Mg are 11 and 12 respectively. Which of the elements has a higher ionization energy? Explain. (2mks)

d) Naturally occurring uranium consists of three isotopes which are radioactive.

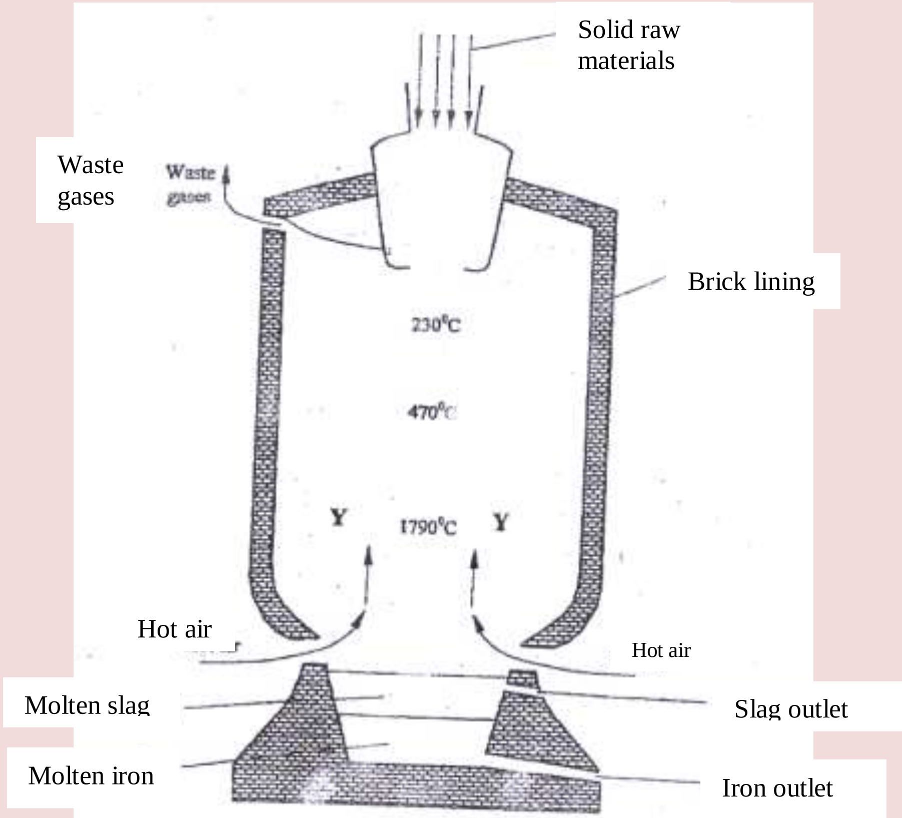
Isotope	^{234}U	^{235}U	^{238}U
Abundance	0.01%	0.72%	99.27%

i) Which of these isotopes has the longest half-life? Give a reason (1 mk)

ii) Calculate the relative atomic mass of uranium (2mks)

iii) $^{235}_{92}U$ Is alpha emitter. If the product of the decay of this nuclide is thorium (Th) . Write a nuclear equation for the process. (1mk)

7. Iron is obtained from haematite using a blast furnace shown if figure 5 below.



a) Four raw materials are required for the production of iron. Three of these are iron oxide, hot air and limestone

Give the name of the fourth raw material (1 mk)

b) Write an equation for the reaction in which carbon (IV) oxide is converted into carbon (II) oxide. (1mk)

c) Explain why the temperature in the region mked Y is higher than that of the incoming hot air. (2 mks)

d) State one physical property of molten slag other than density that allows it to be separated from molten iron as shown in figure 5. (1mk)

e) One of the components of the waste gases is Nitrogen (IV) oxide describe the adverse effects it has on the environment. (2mks)

f) Iron from the blast furnace contains about 5% carbon

i) Describe how the carbon content is reduced. (2mks)

ii) Why is it necessary to reduce the carbon content? (1 mk)

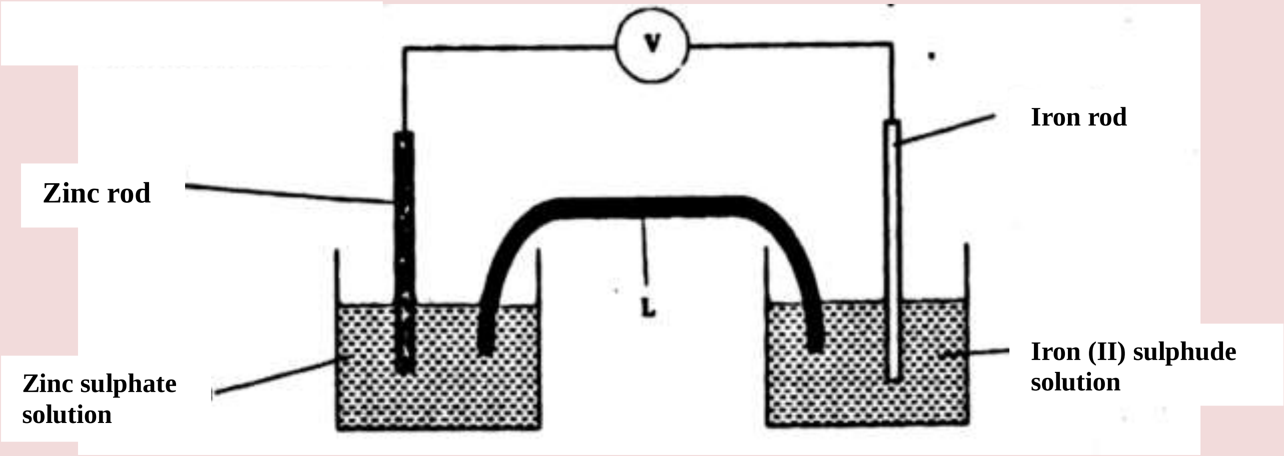
CHEMISTRY PAPER 233/2

K.C.S.E 2010 QUESTIONS

1.

a) Which one of the following compounds; urea, ammonia, sugar and copper (II) chloride will conduct an electric current when dissolved in water? Give reasons. (2mks)

b) The diagram below shows an electrochemical cell. Study it and answer the questions that follows.



Given the following

$Fe^{2+}(aq) + 2e \longrightarrow Fe(s); E^{\theta} = -0.44V$

$Zn^{2+}(aq) + 2e \longrightarrow Zn(s); E^{\theta} = -0.76V$

i) Show on the diagram using an arrow, the direction of flow of electrons (1 mk)

ii) Name **two** subsrances that are used to fill the part labeled L (2 mks)

c) In an experiment to electroplate iron with silver, a current of 0.5 amperes was passed through a solution of silver nitrate for an hour

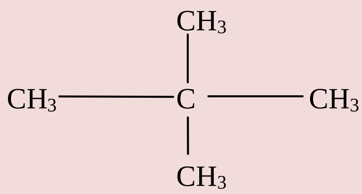
i) Give **two** reasons why it is necessary to electroplate iron with silver (2mks)

ii) Calculate the mass of silver that was deposited on iron (Ag = 108, 1 Faraday = 96,500 coulombs) (3mks)

2.

i. Give the name of the following compounds:

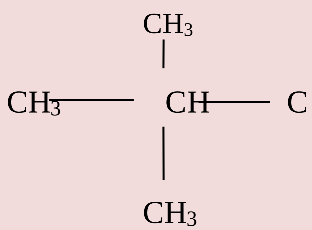
i)



(2 mks)

ii) $CH_3C = CCH_2CH_3$ (1 mk)

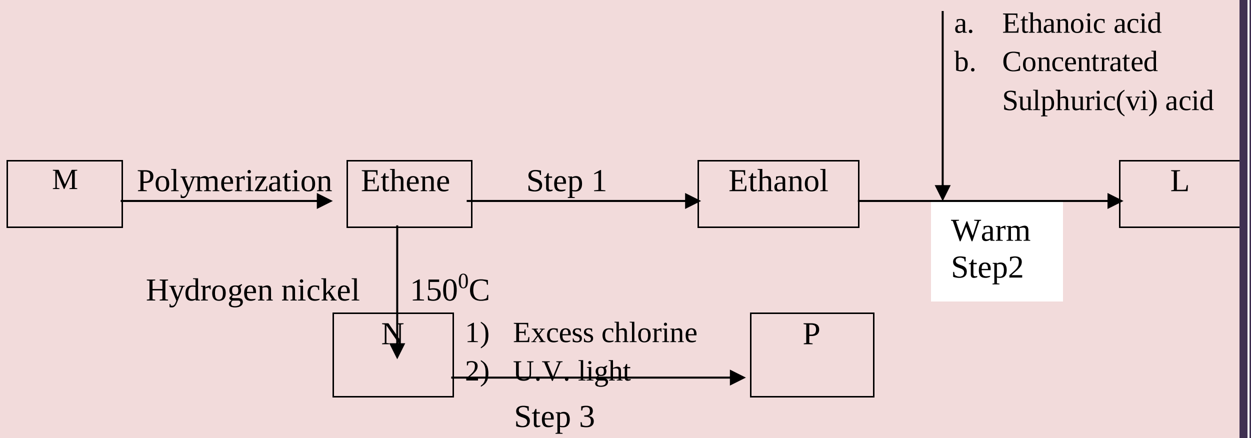
ii. Describe a chemical test that can be carried out in order to distinguish between



and CH_3 and $CH_3C = CCH_2CH_3$

(2 mks)

iii. Study the flow chart below and answer the questions that follows



i) Name the compounds: (2mks)

2. L
3. N

ii) Draw the structural formula of compound M showing two repeat units (1 mk)

iii) Give the reagent and the conditions used in step I (1 mk)

iv) State the type of reaction that take place in: (2mks)

a. Step 2
b. Step 3

2. The molecular formula of compound **P** is $C_2H_2Cl_4$. Draw the two structural formulae

of compound P (2mks)

3. Use the information in the table below to answer the questions that follow. The letters do not represent tha actual symbols of the elements.

Element	Atomic number	Melting point (^0C)
R	11	97.8
S	12	650.0
T	15	44.0
U	17	-102
V	18	-189
W	19	64.0

a) Give the reasons why the meling point of:

i) S is higher than that of R (1 mk)

ii) V is lower than that of U (2mks)

b) How does the reactivity of W with chlorine compare with that of R with chlorine? Explain, (2 mks)

c) Write an equation for the reaction between T and excess oxygen (1 mk)

d) When 1.15g of R were reacted with water, 600cm^3 of gass was produced. Determine the relative atomic mass of R. (Molar gas volume = 24000cm^3) (3mks)

e) Give one use of element V (1 mk)

4.

a. 50cm^3 of 1M copper (II)sulphate solution was placed in a 100cm^3 plastic beaker. The temperature of the solution was measured. Excess metal A powder was added to the solution, the mixture stirred and the maximum temperature was repeated using powder of metals **B** and **C**. The results obtained are given in the table below:

	A	B	C
Maximum temperature (^{0}C)	26.3	31.7	22.0
Initial temperature (^{0}C)	22.0	22.0	22.0

1. Arrange the metal **A, B, C** and copper in order of reactivity starting with the least reactive. Give reasons for the order. (3mks)

2. Other than temperature change, state one other observation that was made when the most reactive metal was added to the copper(II) sulphate solution. (1 mk)

b. The standard enthalpy change of formation of methanol is -239 kJmol^{-1}.

i) Write the thermol chemical equation for the standard enthalpy change of formation of methanol. (1 mk)

ii) Methanol is manufactured by reacting carbon(II)oxide with hydrogen at 300^{0}C and a pressure of 250 atmospheres.
The equation for the reaction is:

$$CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$$

1. How would the yield of methanol be affected if the manufacturing process above is carried out at 300^{0}C and a pressure of 400 atmosphere? Explain (2mks)

2. Use the following data to calculate the enthalpy change for the manufacture of methanol from carbon(II)oxide and hydrogen (3mks)

$$CO_{(g)} + \frac{1}{2} O_{2(g)} \longrightarrow CO_{2(g)}\ ; \quad \Delta H^\theta = -283kJmol^{-1}$$

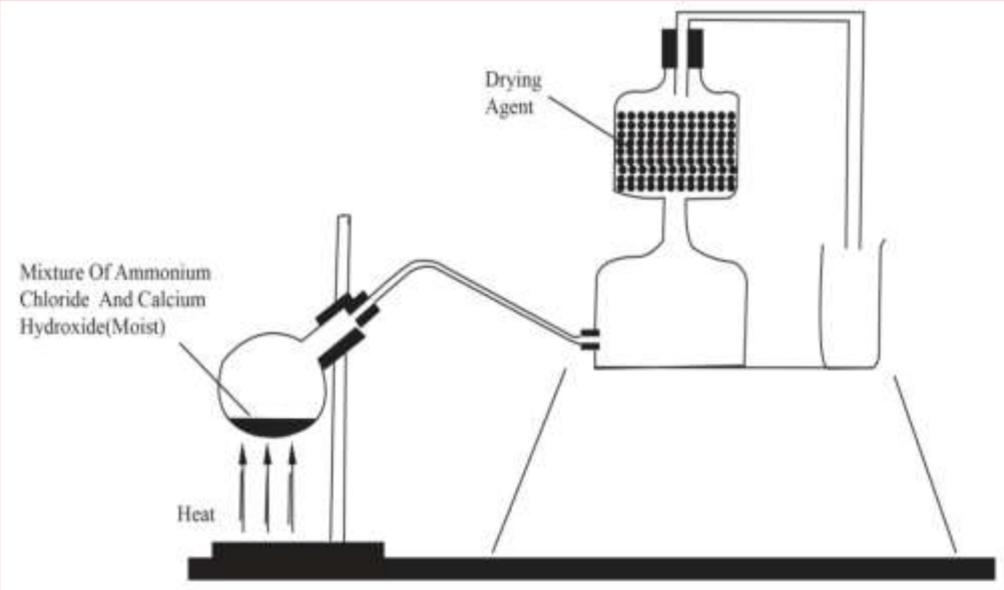
$$H_{2(g)} + \frac{1}{2} O_{2(g)} \longrightarrow H_2O_{(l)}; \quad \Delta H^\theta = -286kJmol^{-1}$$

$$CH_2\ OH_{(l)} + {}^3/_2\ O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(L)}; \quad \Delta H^\theta = -715kJmol^{-1}$$

iii) The calculate enthalpy change in part B(ii) (II) aove differ from the standard enthalpy change of formation of methanol. Give a reason. (1 mk)

5.

a) A student set up the apparatus as shown in the diagram below to prepare and collect dry ammonia gas.



i) Identify **two** mistakes in the set up and give a reason for each mistake. (3mks)

1. Mistake

   Reason

2. Mistake

   Reason

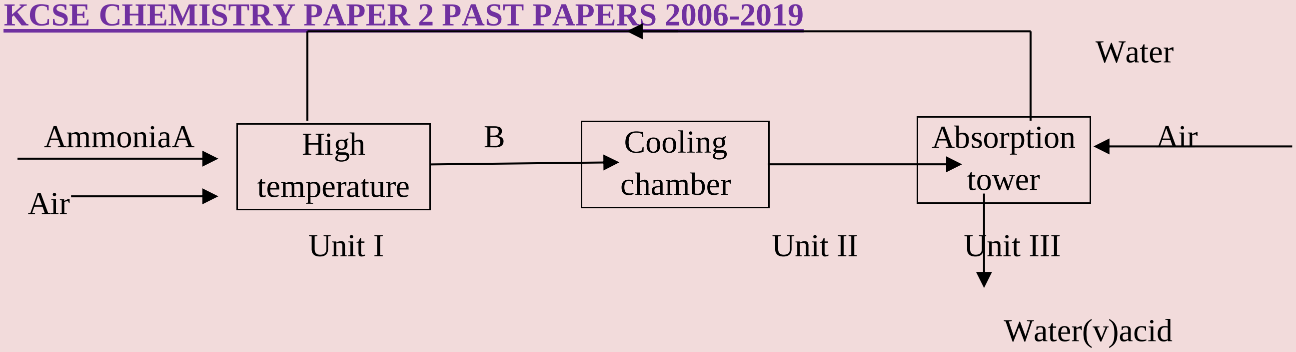
ii) Name a suitable drying agent for ammonia (1 mk)

iii) Write an equation for the reaction that occurred when a mixture of ammonium chloride and calcium hydrogen was heated. (1 mk)

iv) Describe **one** chemical test for ammonia gas (1 mk)

d) Ammonia gas is used to manufacture nitric (V) acid, as shown below.

Gases



i) This process require the use of a catalyst. In which unit is the catalyst used (1 mk)

ii) Identify compound **A** and **B** (1 mk)

iii) Using oxidation number, explain why the conversion of ammonia to nitric(V) acid is called catalytic oxidation of ammonia (2mks)

iv) Ammonia and nitric(V) acid are used in the manufacture of ammonium nitrate fertilizer. Calculate the amount of nitric (V) acid required to manufacture 1000kg ammonium nitrate using excess ammonia. (3 mks)

6. The melting and boiling points of zinc are 419^0C and 907^0C respectively. One of the ores of zinc blende. To extract zinc, the ore is first roasted in air before feeding it into a furnace.

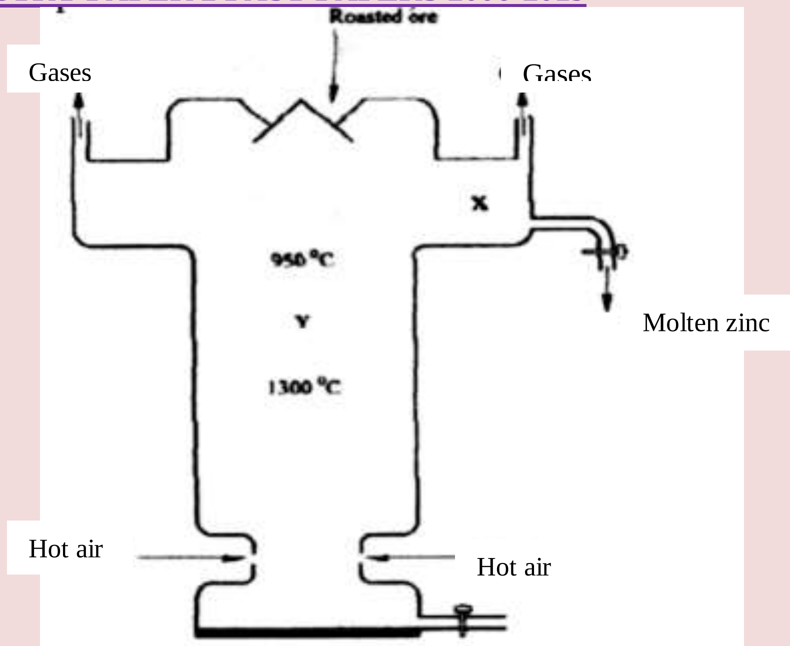
a.

i) Write the formula of the main zinc compound in zinc blende. (1 mk)

ii) Explain using an equation why it is necessary to roast the ore in air before introducing it into the furnace (2 mks)

b. The diagram below shows a simplified furnace used in the extraction of zinc. Study it and answer the questions that follows:

Roasted ore



i) Name **two** other substance that are also introduced into the furnace together with roasted ore. (1 mk)

ii) The main reducing agent in the furnace is carbon(II) oxide. Write **two** equations showing how it is formed. (2 mks)

iii) In which physical state is zinc at point **Y** in the furnace? Give a reason (1 mk)

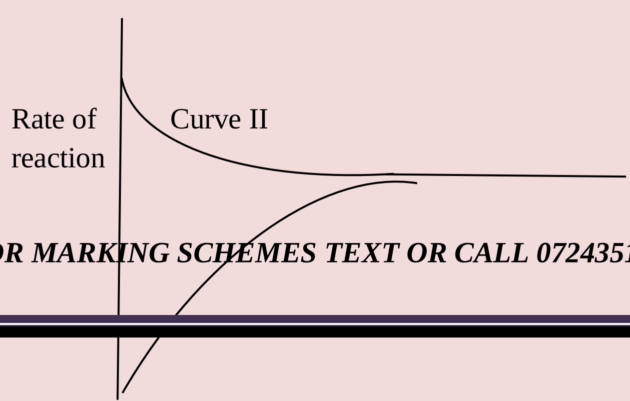
iv) Suggest a value for the temperature at point **X** in the furnace. Give a reason. (1 mk)

v) State and explain **one** environmental effect that may arise from the extraction of zinc from zinc blende (2 mks)

vi) Give **two** industrial uses of zinc. (1 mk)

7. The figure below shows how the rate of the following reaction varies with the time.

$$A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)} + D_{(g)}$$



Y

Curve I

X Time(minutes)

i) Which of the two curves represent the rate of the reverse reaction? Give a reason (2mks)

ii) What is the significance of point **X** and **Y** on the figure? (2mks)

b) State and explain the effect of an increase in pressure on the rates of the following reactions.

i) $H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$ (2mks)

ii) $CH_3OH_{(l)} + CH_3COOH_{(l)} \longrightarrow CH_3COOCH_{3(l)} + H_2O_{(l)}$ (2mks)

c) In an experiment to study the rate of reaction between barium carbonate and dilute hydrochloric acid; 1.97g of barium carbonate were reacted with excess 2M hydrochloric acid. The equation for the reaction is

$$BaCO_{3(s)} + 2HCl_{(aq)} \longrightarrow BaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$$

The data in the table was obtained

Time in seconds	0	30	60	90	120	150	180	210	240
Volume of gas (cm^3)	0	80	135	175	210	230	240	240	240

i) On a grid plot a graph of volume of gas produced (vertical axis) against time (3 mks)

ii) From the graph, determine the rate of the reaction at:

(I) 15 seconds (1 mk)

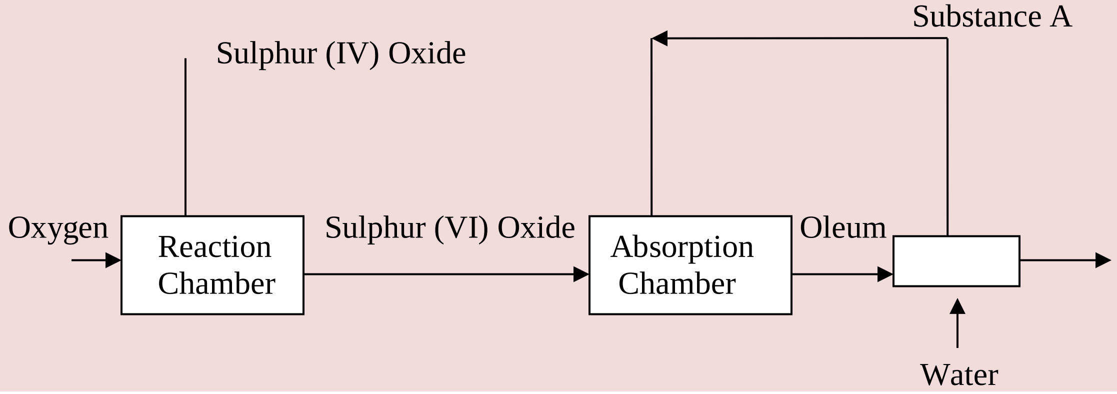
(II) 120 seconds (1 mk)

(III) Give a reason for the difference between the two values. (1 mk)

CHEMISTRY PAPER 233/2

K.C.S.E 2011 QUESTIONS

1. The flow chart below shows some of the processes involved in large scale production of sulphuric (VI) acid. Use it to answer the questions that follow.



a) Describe how oxygen is obtained from air on a large scale (3 mks)

(b) (i) Name substance A.

(ii)Write an equation for the process that takes place in the absorption chamber. (1mk)

(c) Vanadium (V) oxide is a commonly used catalyst in the contact process.

(i) Name another catalyst which can be used for this process. (1 mk)

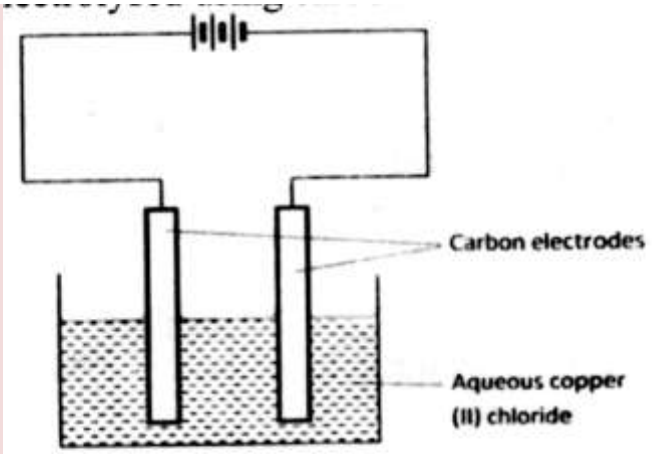
(ii) Give two reasons why vanadium (V) oxide is the commonly used catalyst. (2 mks)

(d) State and explain the observations made when concentrated sulphuric (VI) acid is added to crystals of copper (II) sulphate in a bearer. (2 mks)

(e) The reaction of concentrated sulphuric (VI) acid with sodium chloride produces hydrogen chloride gas. State the property of concentrated sulphuric (VI) acid , illustrated in this reaction. (1mk)

(f) Name four uses of sulphuric (VI) acid

2. The set-up below was used by a student to investigate the products formed when aqueous copper (II) chloride was electrolysed using carbon electrodes.



(a) (i) Write the equation for the reaction that takes place at the cathode. (1 mk)

(ii) Name and describe a chemical test for the product initially formed at the anode when a highly concentrated solution of copper (II) chloride is electrolysed. (3 mks)

(iii) How would the mass of the anode change if the carbon anode was replaced With copper metal? Explain. (2mks)

(b) 0.6 g of metal B were deposited when a current of 0.45A was passed through an electrolyte for 72 minutes. Determine the charge on the ion of metal B.
(Relative atomic mass of B = 59, 1 Faraday = 96 500 coulombs) (3 mks)

(c) The electrode potentials for cadmium and zinc are given below:

$$\mathbf{Cd^{2+}(aq) + 2e \rightleftharpoons Cd(s);\quad E^{\theta} = -0.4v}$$
$$\mathbf{Zn^{2+}(aq) + 2e \rightleftharpoons Zn(s);\quad E^{\theta} = -0.76v}$$

why it is not advisable to store a solution of cadmium nitrate in a container made of zinc (2mks)

3. (a) Ethanol can be manufactured from ethene and steam as shown in the equation below:

$$C_2H_4(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g)$$

Temperature and pressure will affect the position of equilibrium of the above reaction. Name the other factor that will affect the position of equilibrium of the above reaction. (1 mk)

(b) The data in the table below was recorded when one mole of ethene was reacted with excess steam. The amount of ethanol in the equilibrium mixture was recorded under different conditions of temperature and pressure. Use the data to answer the questions that follow.

Temperature (°C)	Pressure (Atm)	Amount of ethanol at equilibrium (Moles)
300	50	0.40
300	60	0.46
300	70	0.55
250	50	0.42
350	50	0.38

(i) State whether the reaction between ethene and steam is exothermic or endothermic. Explain your answer. (3 mks)

(ii) State and explain **one** advantage and one disadvantage of using extremely high pressure in this reaction.

I Advantage
II disadvantage

(c) In an experiment to determine the rate of reaction between calcium carbonate and dilute hydrochloric acid, 2g of calcium carbonate were reacted with excess 2 M hydrochloric acid, The volume of carbon (IV) oxide evolved was recorded at regular intervals of one minute for six minutes. The results are shown in the table below.

Time (minutes)	1	2	3	4	5	6
Volume of carbon (IV) oxide (cm^3)	170	296	405	465	480	480

(i) plot a graph of time in minutes on the horizontal axis against volume of carbon (IV) oxide on the vertical axis.

(ii) determine the rate of reaction at 4 minutes (2mks)

4. (a) When excess calcium metal was added to 50 cm^3 of 2 M aqueous copper (II) nitrate in a beaker, a brown solid and bubbles of gas were observed.

(i) "Write two equations for the reactions which occurred in the beaker. (2 mks)

(ii) Explain why it is not advisable to use sodium metal for this reaction.

(b) Calculate the mass of calcium metal which reacted with copper (II) nitrate solution. (Relative atomic mass of Ca = 40) (2 mks)

(c) The resulting mixture in (a) above was filtered and sodium hydroxide added to the filtrate dropwise until in excess. What observations were made? (1mk)

(d) (i) Starting with calcium oxide, describe how a solid sample of calcium carbonate can be prepared.

(ii) Name one use of calcium carbonate

5. (a) Other than their location in the atom, name two other differences between an electron and a proton.

(b) the table below gives the number of electrons ,protons and neutrons in particles A,B, C , D, E, F and G

particle	Protons	electrons	neutrons
A	6	6	6
B	10	10	12
C	12	10	12
D	6	6	8
E	13	10	14
F	17	17	18
G	8	10	8

(i) Which particle is likely to be a halogen? (1 mk)

(ii) what is the mass number of E

(iii) write the formula of the compound formed when E combines with G (1 mk)

iv) Name the type of bond formed in (iii) above.

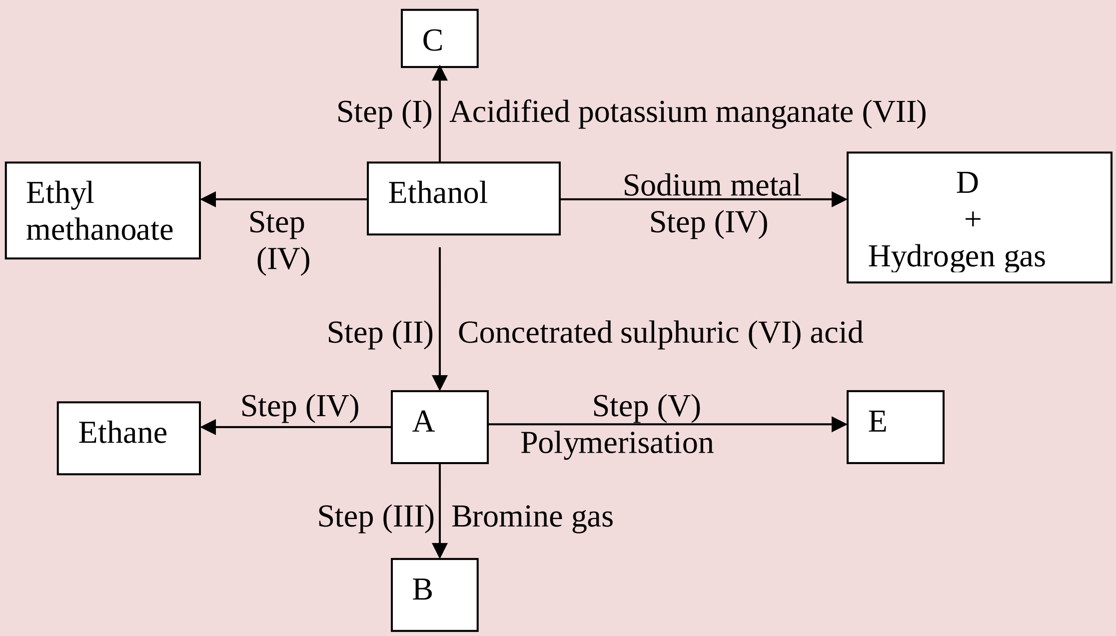
(v) How does the radii of C and E compare ? Give reason. (2mks)

(vi) Draw a dot (.) and cross(x) diagram for the compound formed between

(1mk)

(vii) Why would particle B not react with particle D? (1mk)

5.



(i) I What observation will be made in Step *I* (1 mk)

II Describe a chemical test that can be carried out to show the identity of compound C (2 mk)

(ii)Give the names of the following

I **E**………………………………………………………………… (2 mk)

II substance **D** ………………………………………………………… (1 mk)

(iii)Give the formula of substance **B .**

(iv) Name the type of reaction that occurs in:

I Step (II)

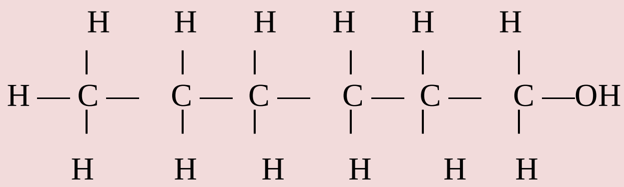
II Step (IV)

(v) Give the reagent and conditions necessary for Step (VI).

Rea2ent:

Conditions

(b) (i) Name the following structure.



(ii) Draw the structure of an isomer of pentene.

7. (a) What is meant by molar heat of combustion? (1 mk)

(b) State the Hess's Law.

c) Use the following standard enthalpies of combustion of graphite, hydrogen and enthalpy of formation of propane.

ΔH^{θ}_{c} (Graphite) = -393 Kj mol^{-1}

ΔH^{θ}_{c} (H_2(g)) = -286Kj mol^{-1}

ΔH^{θ}_{f} (C_3H_8 (g) = -104 Kj mol^{-1}

(i) Write the equation for the formation of propane. (1 mk)

(ii) Draw an energy cycle diagram that links the heat of formation of propane with its heat of combustion and the heats of combustion of graphite and hydrogen.

(iii) Calculate the standard heat of combustion of propane. (2 mks)

Other than the enthalpy of combustion, state **one** factor which should be considered when choosing a fuel. (1 mk)

(e) The molar enthalpies of neutralization for dilute hydrochloric acid and dilute nitric (V) acid are -57.2kJ/mol while that of ethanoic acid is -55.2kJ/mol. Explain this. observation. (2mks)

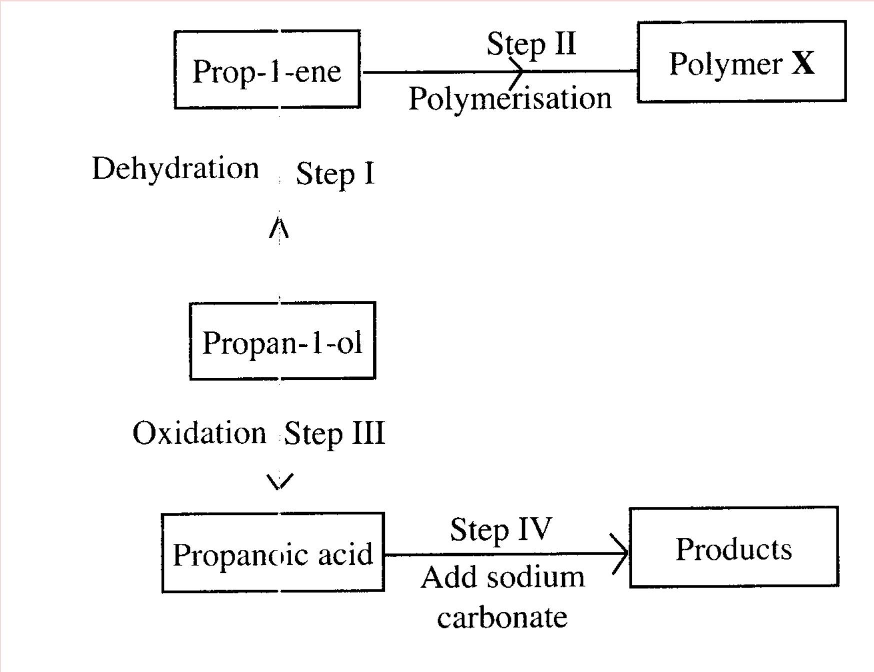
CHEMISTRY PAPER 233/2

K.C.S.E 2012 QUESTIONS

1. a) Draw the structural formula for all the isomers of $C_2H_3CL_3$ (2mks)

b) Describe two chemical tests that can be used to distinguish between ethane and ethane. (4mks)

c) The following scheme represents various reactions starting with propan-i-ol. Use it to answer the questions that follow.



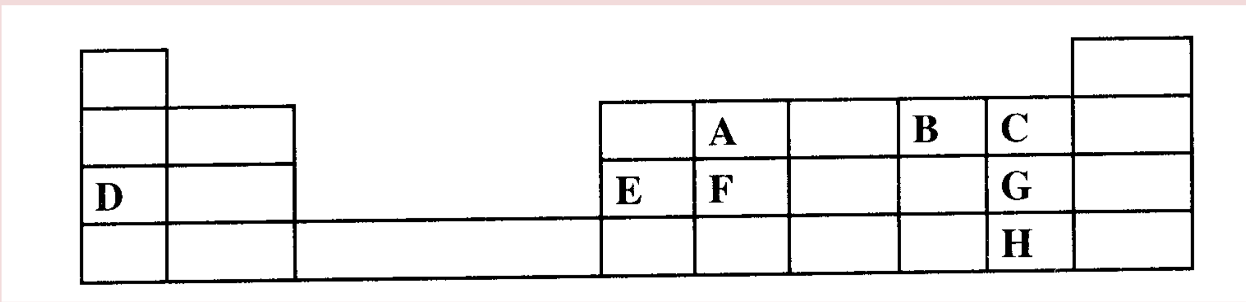
i) Name one substance that can be used in step I. (1mk)

ii) Give the general formula of X. (1 mk)

iii) Write the equation for the reaction in step IV. (1mk)

iv) Calculate the mass of propan-I-ol which when burnt completely in air at room temperature and pressure would produce $18dm^3$ of gas. (C = 12.0; O = 16.0; H = 1.0; Molar gas volume = $24dm^3$) (3mks)

2. The grid below is part of the periodic table. Use it to answer the questions that follow. (The letters are not the actual symbols of the elements.)



a) Which is the most reactive non-metallic element shown in the table? Explain. (2mks)

b) i) Write the formula of the compound formed when element A reacts with element B. (1mk)

ii) Name the bond type in the compound formed in b (i) above. (1mk)

c) i) What is the name given to the group of elements where C, G and H belong? (1mk)

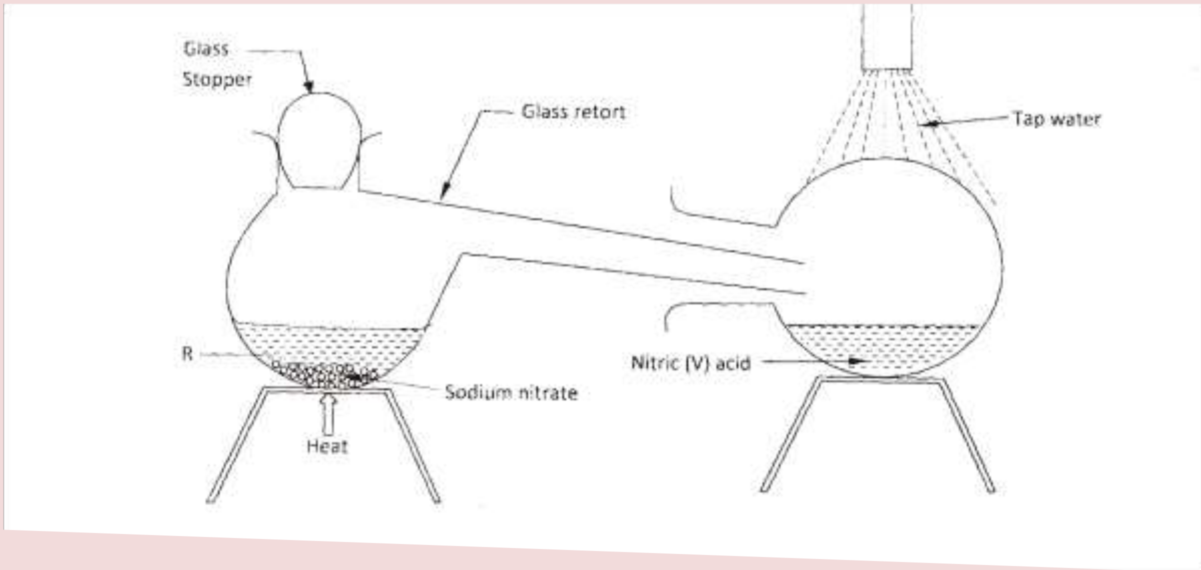
ii) Write an equation for the reaction that occurs when C in gaseous form is passed through a solution containing ions of element H. (2mks)

d) The melting points of elements F and G are 1410oC and -101^{o}C respectively. In terms of structure and bonding, explain why there is a large difference in the melting points of F and G. (2mks)

e) D forms two oxides. Write the formula of each of the two oxides. (1mk)

f) J is an element that belongs to the 3rd period of the periodic table and a member of the alkaline earth elements. Show the position of J in the grid. (1mk)

3. In the laboratory, small quantities of nitric (V) acid can be generated using the following set up. Study it and answer the questions that follow.



a) i) Give the name of substance R. (1mk)

ii) Name one other substance that can be used in place of sodium nitrate. (1mk)

iii) What is the purpose of using tap water in the set up above? (1mk)

b) Explain the following;

i) It is not advisable to use a stopper made of rubber in the set-up (1mk)

ii) the reaction between copper metal with 50% nitric (V) acid in an open test-tube produces brown fumes. (1mk)

c) i) Nitrogen is one of the reactants used in the production of ammonia, name two sources of the other reactant. (2mks)

ii) A factory uses nitric (V) acid and ammonia gas in the preparation of a fertilizer. If the daily production of the fertilizer is 4800kg; calculate the mass of ammonia gas used in kg. (N = 14.0; O = 16.0; H = 1.0) (3mks)

iii) State two other uses of nitric (V) acid other than the production of fertilizers. (2mks)

4. The factors which affect the rate of reaction between lead carbonate and dilute nitric (V) acid were investigated by carrying out three experiments;

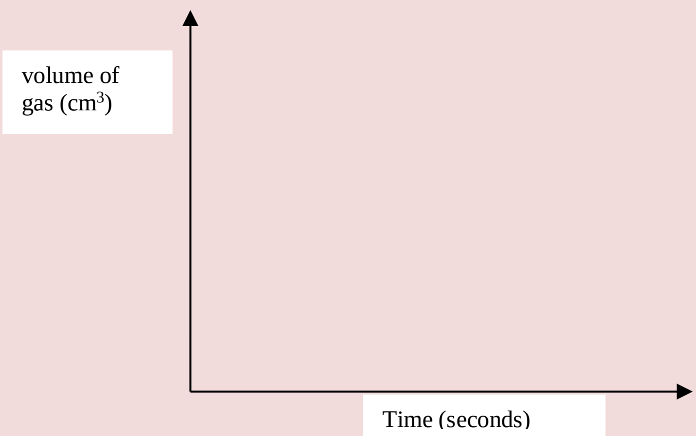
Experiment number	Lead carbonate	Concentration of nitric (V) acid
1	Lumps	4M
2	Powdered	4M
3	Lumps	2M

a) Other than concentration , name the factor that was investigated in the experiments. (1mk)

b) For each experiment, the same volume of acid (excess) and mass of lead carbonate were used and the volume of gas liberated measured with time.

i) Draw a set up that can be used to investigate the rate of reaction for one of the experiments. (3mks)

ii) On the grid provided, sketch the curves obtained when the volume of gas produced was plotted against time for each of the three experiments and label each as 1, 2 or 3. (4mks)



iii) Write an equation for the reaction that took place. (1mk)

c) If the experiments were carried out using dilute hydrochloric acid in place of dilute nitric (V) acid, the reaction would start, slow down and eventually stop. Explain these observations. (2mks)

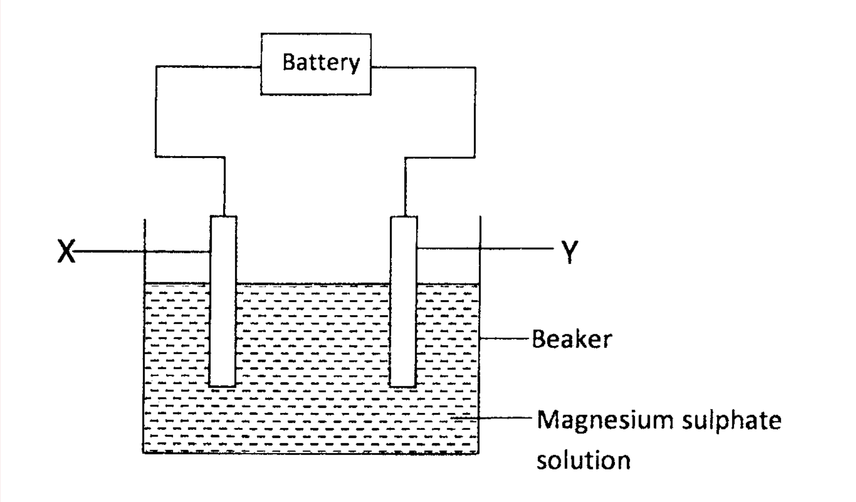
d) A solution of bromine gas in water is an example of a chemical reaction in a state of balance. The reaction involved is represented by the equation below.

$$Br_{2(g)} + H_2O_{(L)} \rightleftharpoons 2H^+_{(aq)} + Br^-_{(aq)} + OB^-_{(aq)}$$

Yellow/orange Colourless

State and explain the observation made when hydrochloric acid is added to the mixture at equilibrium. (2mks)

5. a) The set up below was used to investigate the products formed at electrodes during electrolysis of aqueous magnesium sulphate using inert electrodes. Use it to answer the questions that follow.



i) During the electrolysis, hydrogen gas was formed at electrode Y. identify the anode. Give a reason for your answer. (2mks)

ii) Write the equation for the reaction with takes place at electrode X. (1mk)

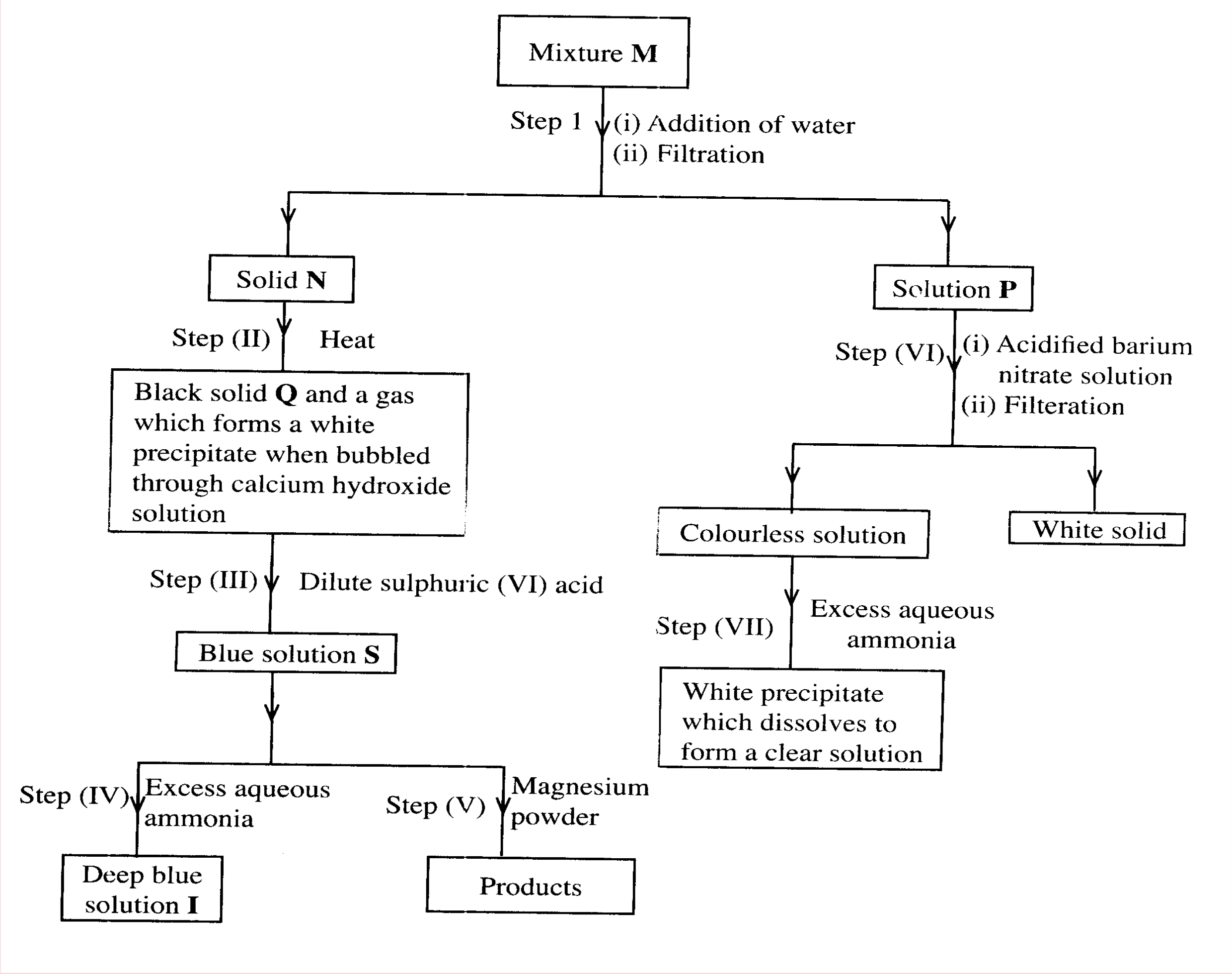
iii) Why is the concentration of magnesium sulphate expected to increase during electrolysis? (2mks)

iv) What will be observed if red and blue litmus papers were dipped into the solution after electrolysis? (2mks)

b) During electrolysis of magnesium sulphate, a current of 0.3A was passed for 30 minutes. Calculate the volume of gas produced at the anode. (Molar gas volume = $24dm^3$; 1 Faraday = 96,500C.). (3mks)

c) State two applications of electrolysis. (1mk)

6. The flow chart below shows a sequence of reactions involving a mixture of two salts, mixture M. Study it and answer the questions that follow.



a) Write the formula of the following;

i) anion in solid Q (1mk)

ii) the two salts present in mixture M. (2mks)

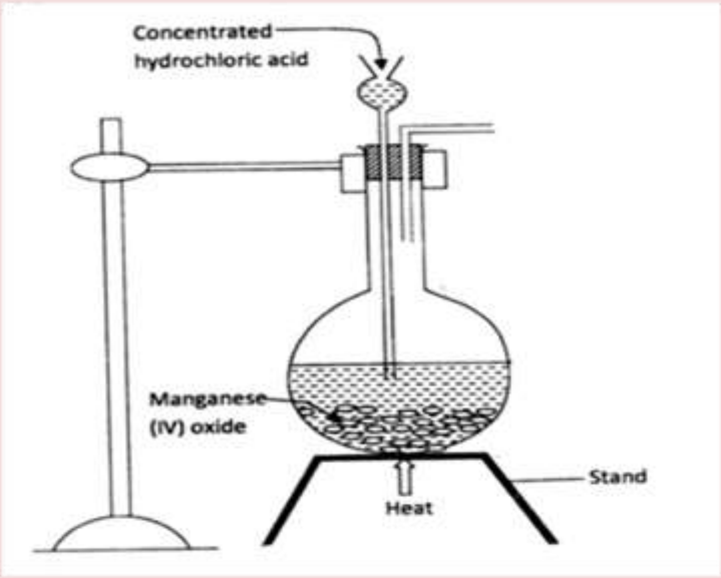
b) Write an ionic equation for the reaction in step (VI) (1mk)

c) State and explain the observations made in step (V). (3mks)

d) i) Starting with Lead (II) oxide, describe how a pure solid sample of lead sulphate can be prepared in the laboratory. (2mks)

ii) How can one determine whether the lead sulphate prepared is pure? (2mks)

7. a) The diagram below is part of set up used to prepare and collect dry chlorine gas.



i) Complete the diagram to show how a dry sample of chlorine gas can be collected. (3mks)

ii) Name another substance and condition that can be used instead of manganese (VI) oxide. (1mk)

iii) Write an equation for each of the following;

I. chlorine gas reacting with iron (1 mk)

II. chlorine gas reacting with hot concentrated sodium hydroxide solution. (1mk)

b) An oxide of chlorine of mass 1.83g was found to contain 1.12g of oxygen. Determine the empirical formula of the oxide (O = 16.0; Cl = 35.5). (3mks)

c) Other than the manufacture of weed killers, name two other uses of chlorine. (2mks)

CHEMISTRY PAPER 233/2

K.C.S.E 2013 QUESTIONS

1. The grid given below represents part of the periodic table. Study it and answer the questions that follow. The letters do not represent the actual symbol of the element

M				N	P	T		
R								

(i) Select a letter which represents an element that looses electrons most readily. Give a reason for your answer. (2mks)

(ii) Explain why the atomic radius of **P** is found to be smaller than that of **N** (2mks)

(iii) Element **M** reacts with water at room temperature to produce 0.2 dm^3 of gas. Determine the mass of **M** which was reacted with water. (molar gas volume at room temperature is 24 dm^3 , relative atomic mass of M=7 (3mks)

(b) Use the information in the table below to answer the question that follows. (The letters are not the symbols of the elements)

Element	State of oxide at room temperature	Type of oxide	Bonding in oxide
U	Solid	Acidic	Covalent
W	Solid	Basic	Ionic
X	Liquid	Neutral	Covalent
Y	gas	neutral	covalent

Identify a letter which represents an element in the table that could be calcium, carbon or sculpture. Give reasons in each case.

(i) Calcium: (2mks)

Reason

(ii) Carbon (2mks)

Reason

(iii)Sulphur: (2mks)

Reason

2. (a) (i) what is meant by the term Enthalpy of formation? (1mk)

(ii) The enthalpies of combustion of carbon, methane and hydrogen are indicated below:

$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)};\ \Delta H = -393\ kJ\ mol^{-1}$

$H_{2(g)} + \frac{1}{2} O_{2(g)} \longrightarrow H_2O_{(l)};\ \Delta = -286\ kJ\ mol^{-1}$

Enthalpy of combustion of $CH_4 = -890\ kJ\ mol^{-1}$

(i) Draw an energy cycle diagram that links the enthalpy of formation of methane to enthalpies of combustion of carbon, hydrogen and methane (2mks)

(ii) Determine the enthalpy of formation of methane (2mks)

(b) An experiment was carried out where different volumes of dilute hydrochloric acid and aqueous sodium hydroxide both at 250c were mixed and stirred with a thermometer. The highest temperature reached by each mixture was recorded in the table below

Volume of hydrochloric acid (cm^3)	5	10	15	20	25	30	35	40	45
Volume of sodium hydroxide (cm^3)	45	40	35	30	25	20	15	10	5
Highest temperature of mixture(^oc)	27.2	29.4	31.6	33.8	33.6	31.8	30.0	28.4	26.6

(i) On the grid provided. plot a graph of highest temperature (vertical axis) against volume of hydrochloric acid (horizontal axis) (3mks)

(ii) Using your graph , determine the

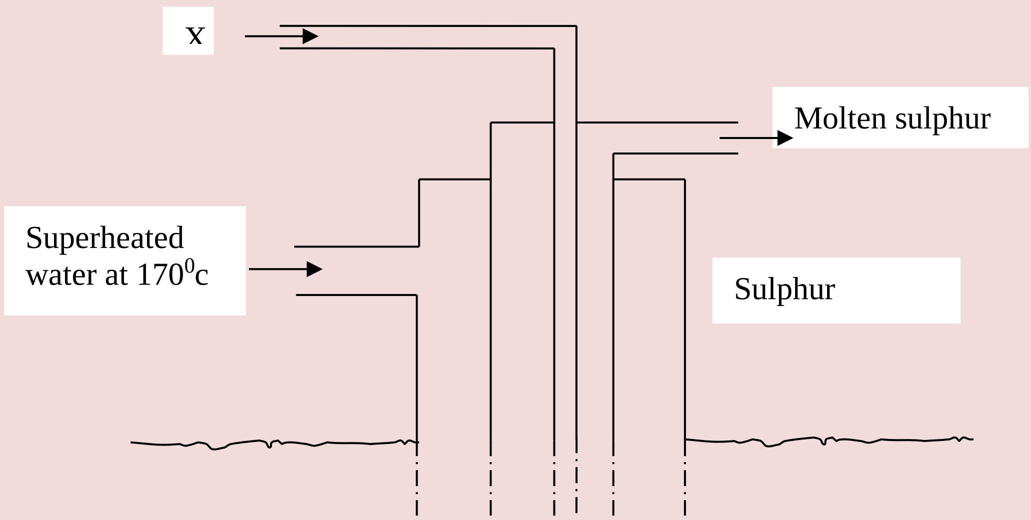
(a) Highest temperature reached ($^1/_2$mk)

(b) Volume of acid and base reacting when highest temperature is reached ($^1/_2$mk)

(iii) Calculate the amount of heat liberated during the neutralization process .(specify heat capacity is 4.2 j g $^{-1}$ k^{-1} and the density of solution is 1.0 g cm^{-3}) (2mks)

(c). The molar enthalpy of neutralization between hydrochloric acid and ammonia solution was found to be -52.2 kJ mol^{-1}, while that of hydrochloric acid and sodium hydroxide was – 57.1 Kj mol^{-1}. Explain the difference in these values. (2mks)

3. (a) The diagram below shows the frasch process used for extraction of sulphur Use it to answer the question that follows

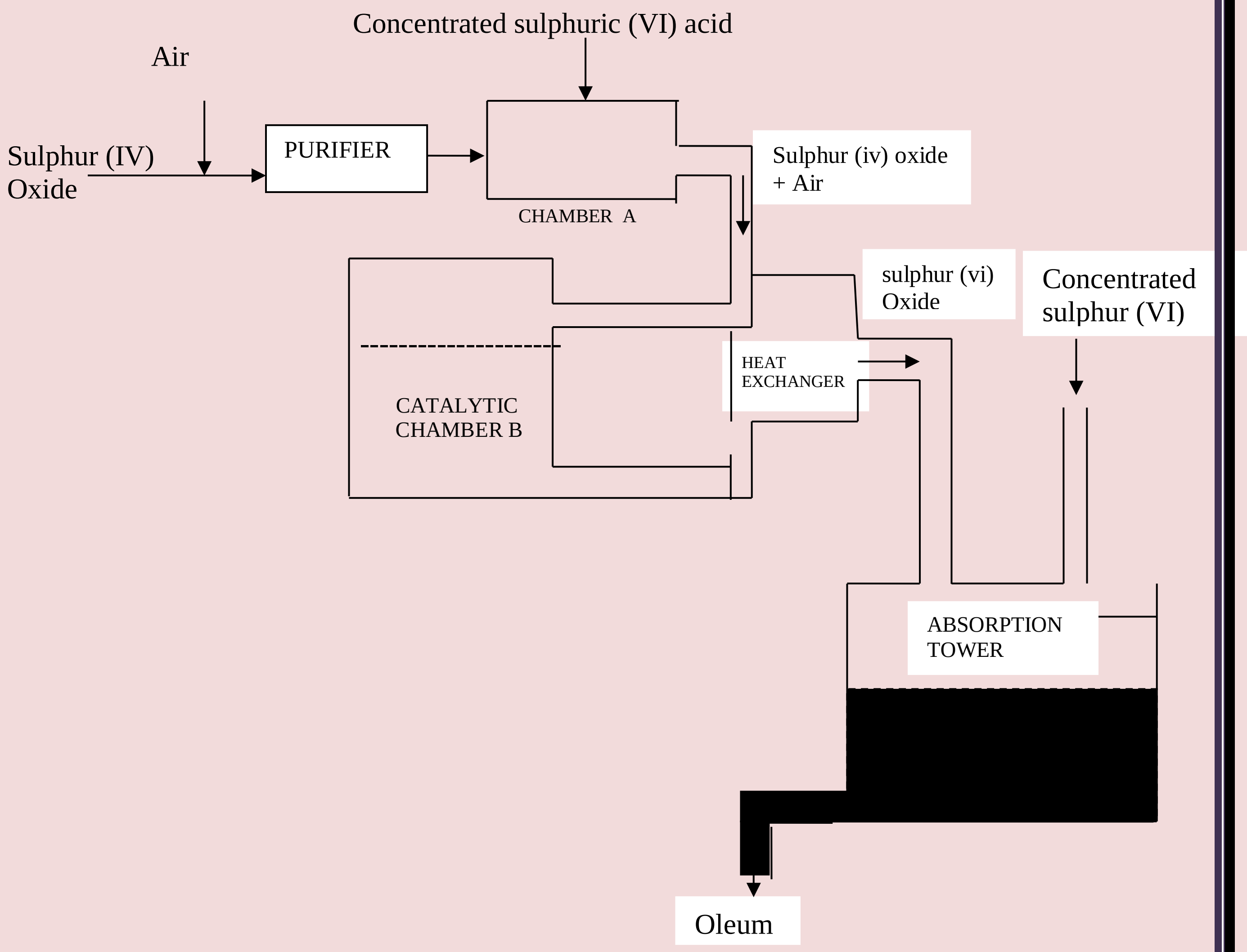


(i) Identify X (1mk)

(ii) Why is it necessary to use superheated water in this process (1mk)

(iii) State two physical properties of sulphur that makes it possible for it to be extracted by this method (2mks)

(b) The diagram below shows part of the process in the manufacture of sulphuric (VI) acid. Study it and answer the questions that follow



(I) Write an equation for the formation of sulphur (IV) oxide from sulphur (1mk)

(II) What is the role of concentrated sulphur (VI) acid in chamber A? (1 mk)

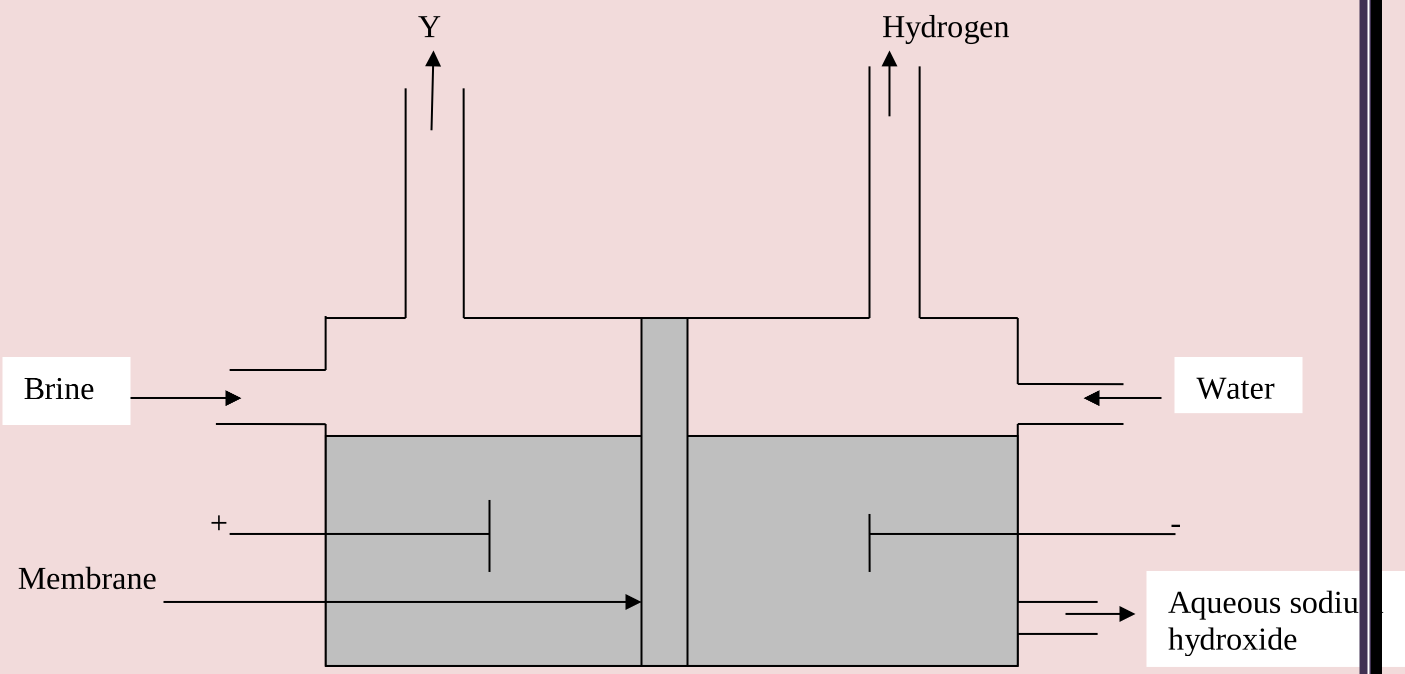
(III) Name two catalysts that can be used in the catalytic chamber B. (2 mks)

(IV) State two roles of the heat exchanger (1 mk)

(c) Explain one way in which sulphur (IV) oxide is a pollutant (1mk)

(d).what observation will be made when a few drops of concentrated sulphuric (VI) acid are added to crystals of sugar? Explain your answer (1 mk)

4. (a) the set below can be used to produce sodium hydroxide by electrolyzing brine



(i) Identify gas Y (1 mk)

(ii) Describe how aqueous sodium hydroxide is formed in the above set-up (2mks)

(iii)One of the uses of sodium hydroxide is in the manufacturing of soaps state one other use of sodium hydroxide (1 mk)

(b).study the information given in the table below and answer the question that follows

Half reaction	Electrode potential E^0V

$D^{2+}_{(aq)} + 2e \longrightarrow D_s$	-0.13
$E^{+}_{(aq)} + e \longrightarrow E_s$	+0.80
$F^{3+}_{(aq)} + e \longrightarrow F^{2+}_{aq}$	+0.68
$G^{2+}_{(aq)} + 2e \longrightarrow G_s$	-2.87
$H^{2+}_{(aq)} + 2e \longrightarrow H_s$	+0.34
$J^{+}_{(aq)} + e \longrightarrow J_{(s)}$	-2.71

(i) Construct an electrochemical cell that will produce the largest emf (3mks)

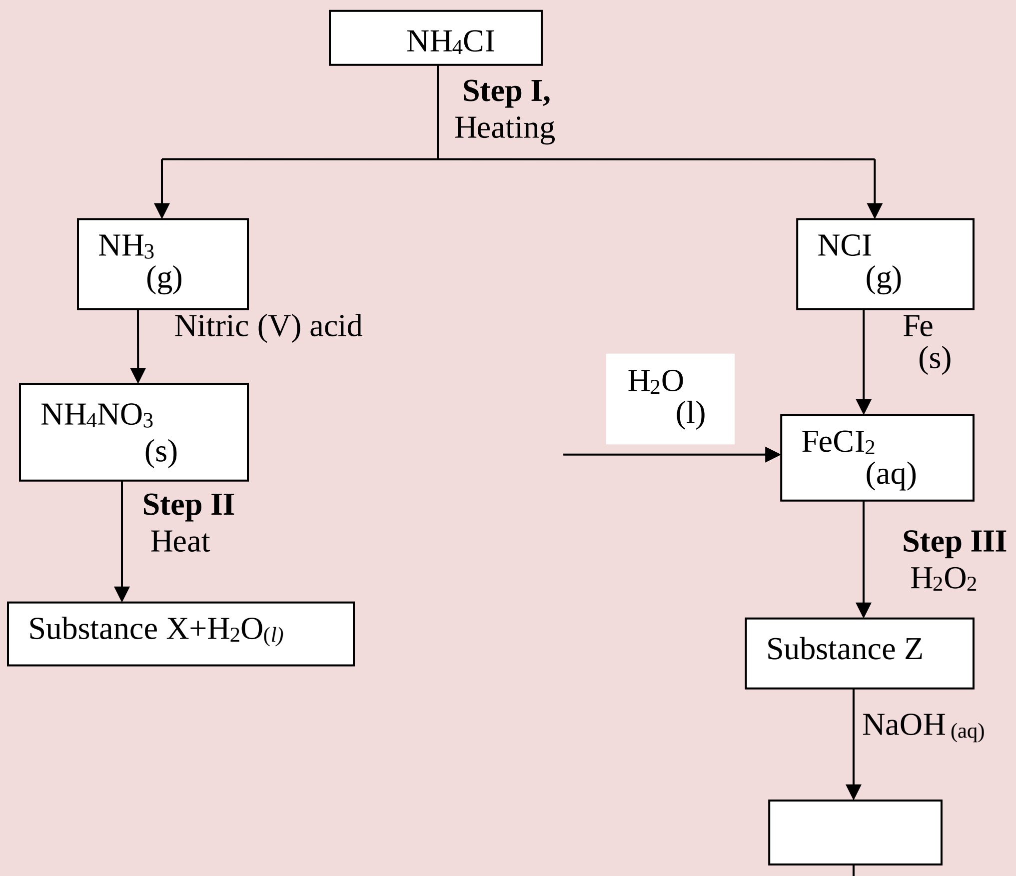
(ii) Calculate the emf of the cell constructed in(i) above (2mks)

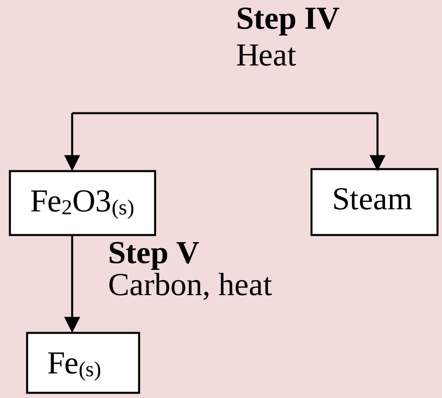
(iii) Why is it not advisable to store a solution containing E^{+} ions in a container made of **H**? (2mks)

5. (a) describes one method that can be used to distinguish between sodium sulphate and sodium hydrogen sulphate. (2mks)

b) Describe how a pure sample of lead (II) sulphate can be prepared in the laboratory starting with lead metal (3mks)

c) Study the flow chart below and answer the questions that follow:





i) Write an equation for the reaction in:

I step II; (1 mk)

II step IV (1 mk)

ii) State the observation made in step III. Explain. (2mks)

iii) Name another substance that can be used in step V. (1 mk)

6. (a) distinguish between a neutron and proton (1 mk)

(b) What is meant by a radioactive substance? (1 mk)

(c) State two dangers associated with radioactive substance in the environment (2mks)

(d) The two isotopes of hydrogen, deuterium $^{2}_{1}D$ ant tritium $^{3}_{1}T$ react to form element y and neutron particles, according to the equation below

$$^{2}_{1}D + ^{3}_{1}T \longrightarrow ^{a}_{b}t + ^{1}_{1}n$$

(i) What is the atomic

a. Mass of Y (1 mk)

b. Number of Y (1 mk)

(ii) What name is given to the type of reaction undergone by the isotopes of hydrogen? (1 mk)

(e).(i) What is meant by half-life of a radioactive substance (1 mk)

(ii) 288 g of a radioactive substance decayed to 9 g in 40 days. Determine the half-life of the radioactive substance (2mks)

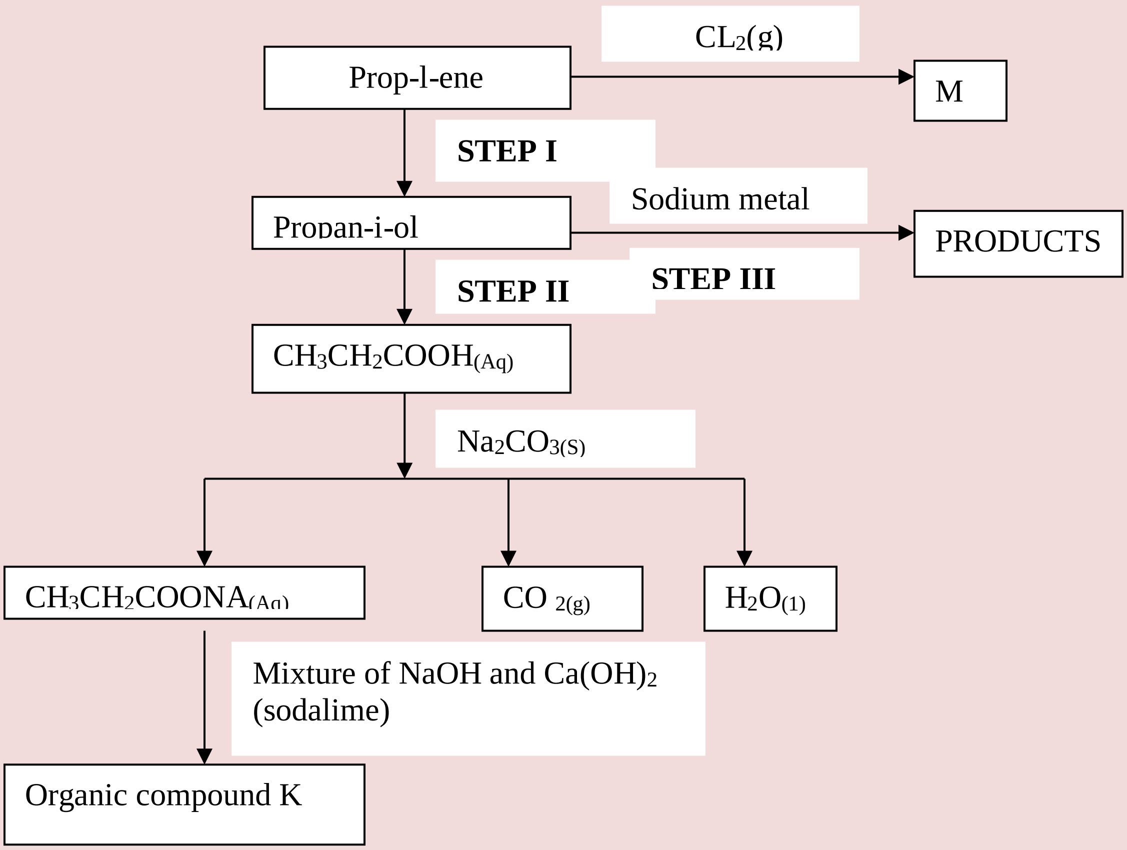
7.(a) Give the systematic names for the following compounds

(i) CH_3CH_2COOH; (1mk)

(ii) $CH_3CH_2CH_2CHCH_2$; (1mk)

(iii) $CHC\ CH_2CH_3$; (1mk)

(b) Study the flow chart below and use it to answer the questions that follow



(i) Identify the organic compound K (I mk)

(ii) Write the formula of M (1 mk)

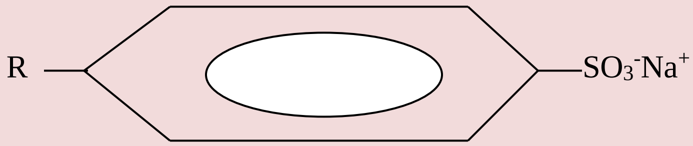
(iii) Give one reagent that can be used in

(a) Step I (1mk)

(b) Step II (1 mk)

(iv) Write the equation of the reaction in step III (1 mk)

(c) The structure below represents a type of a cleaning agent

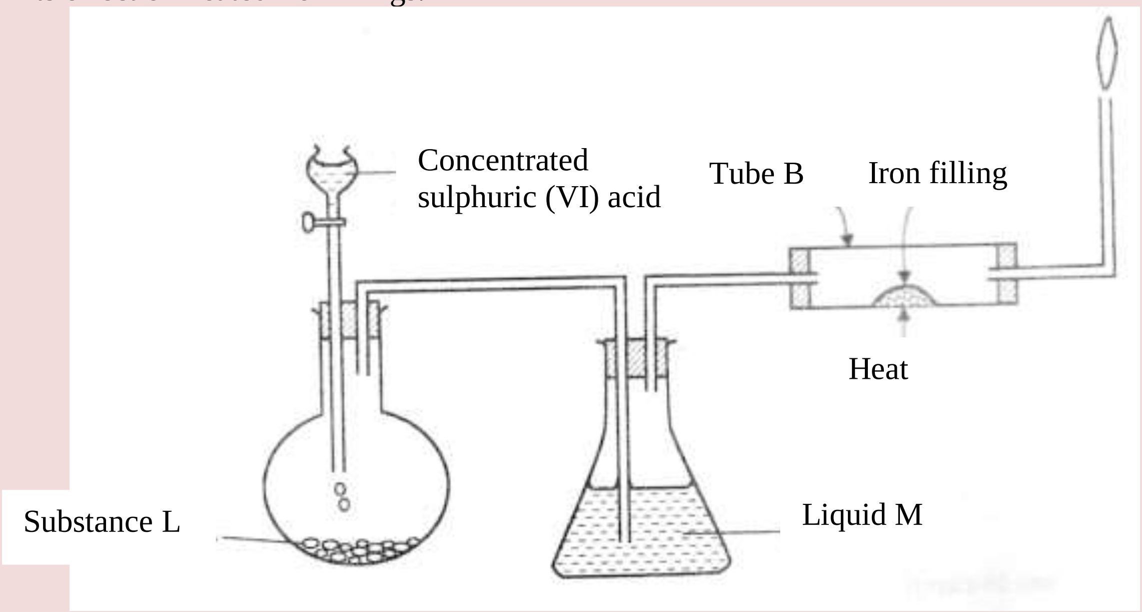


Describe how the cleansing agent removes grease from a piece of cloth (3mks)

CHEMISTRY PAPER 233/2

K.C.S.E 2014 QUESTIONS

1. (a) The set up below was used to prepare dry hydrogen chloride gas, and investigate its effect on heated iron filings.



(i). Name substance **L** (1 mk)

(ii) Name liquid **M** (1 mk)

(iii) What will be observed in tube **B**? (1 mk)

(iv) Write an equation for the reaction that occurs in tube **B**. (1 mk)

(v) Why is the gas from tube **B** burnt? (1 mk)

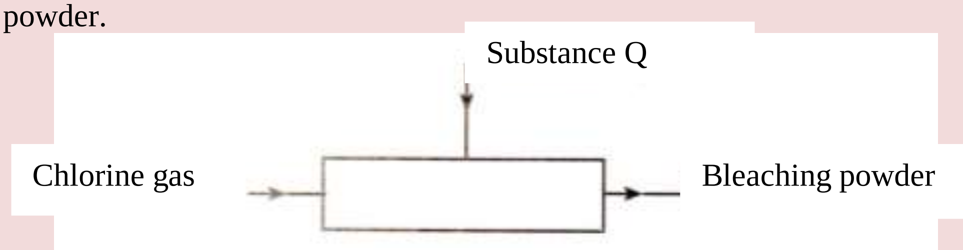
(b) (i) Explain the following observations:

I) A white precipitate is formed when hydrogen chloride gas is passed through aqueous silver nitrate. (1 mk)

II) Hydrogen chloride gas fumes in ammonia gas. (1 mk)

(ii) State **two** uses of hydrogen chloride gas (1 mk)

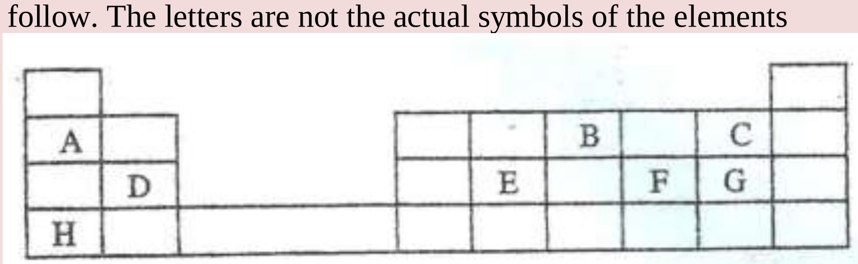
(c) The diagram below is a representation of an industrial process for the manufacture of a bleaching powder.



(i) Name substance **Q.** (1 mk)

(ii)When the bleaching powder is added to water during washing, a lot of soap is used. Explain (1 mk)

2. (a) The grid below represents part of the periodic table . Study it and answer the questions that follow. The letters are not the actual symbols of the elements



i) Select the most reactive metal. Explain (2 mks)

ii) Select an element that can form an ion with a charge of 3^- (1 mk)

iii) Select an alkaline earth metal (1 mk)

iv) Which group 1 element has the highest first ionization energy? Explain (2 mks)

v) Element A combines with chlorine to form a chloride of A. State the most likely pH value of a solution of a chloride of A. Explain (2 mks)

(b) (i) Explain why molten calcium chloride and magnesium chloride conduct electricity while carbon tetrachloride and silicon tetrachloride do not. (2 mks)

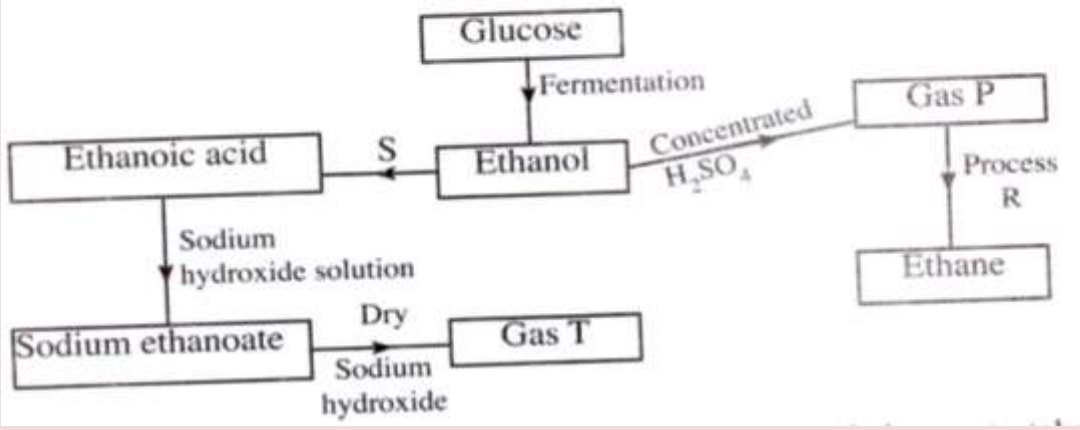
(ii) Under the same conditions , gaseous neon was found to diffuse faster than gaseous fluorine. Explain this observation. (F=19.0;Ne=20.0) (2 mks)

3. (a) Draw the structures of the following.

(i)Butan -1-*0l* (1 mk)

(ii)Hexanoic acid. (1 mk)

(b) Study the flow chart below and answer the questions that follow



(i) State the conditions necessary for fermentation of glucose to take place (1 mk)

(ii) State **one** reagent that can be used to carry out process S. (1 mk)

(iii) Identify gases (2 mks)

P:

T:

(iv) How is sodium hydroxide kept dry during the reaction (1 mk)

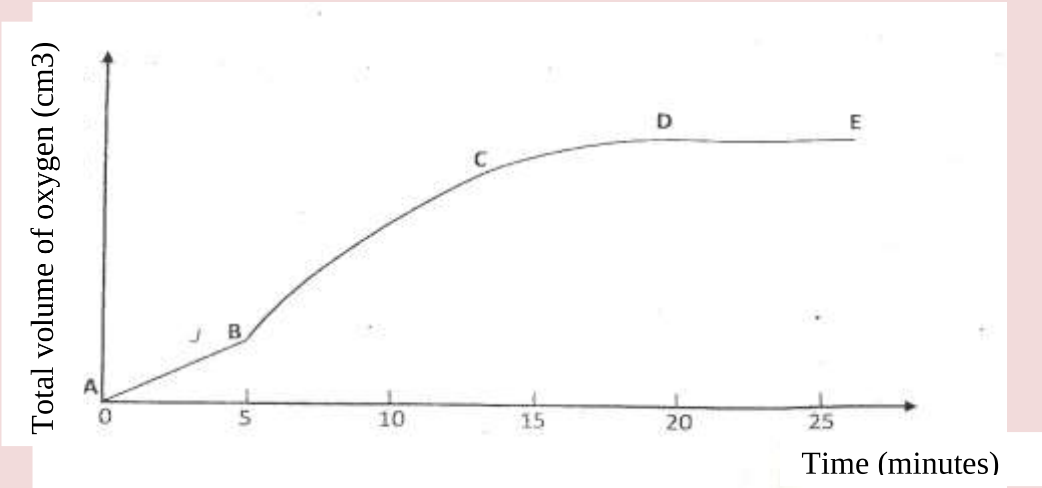
(v) Give **one** commercial use of process **R.** (1 mk)

(c) When one mole of ethanol is completely burnt in air, 1370kJ of heat energy is released. Given that 1 lire of ethanol is 780 g , calculate the amount of heat energy released when 1 litre of ethanol is completely burnt
(C = 12.0; H=1.0; 0=16.0) (3 mks)

(d) State **two** uses of ethanol other than as an alcoholic drink. (2 mks)

4. (a) Other than temperature, state **two** factors that determine the rate of a chemical reaction. (2mks)

(b) A solution of hydrogen peroxide was allowed to decompose and the oxygen gas given off collected. After 5 minute, substance **G** was added to the solution of hydrogen peroxide. The total volume of oxygen evolved was plotted against time as shown in the graph below



(i)Describe the procedure of determining the rate of the reaction at minute 12. (3mks)

(ii)How does the production of oxygen in region AB compare with that in region BC? Explain (2 mks) (iii)Write an equation to show the decomposition of hydrogen peroxide. (1 mk)

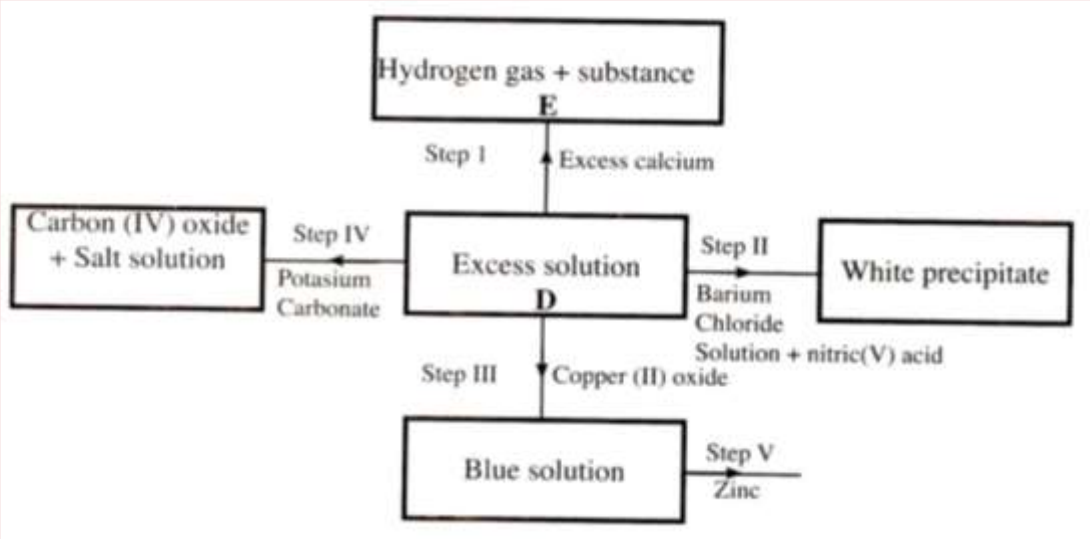
(c) Sulphur (IV) oxide react with oxygen to form Suplhur (VI) oxide as shown in the equation below

$2SO_{2(g)} + O_{2(g)} \leftrightarrows 2SO_{3\,(g)}$: $\Delta H = {}^{-}192kJ$:

(i)Explain the effect on the yield of SO_3 of lowering the temperature of this reaction. (2 mks)

(ii)Name **one** catalyst used for the reaction. (1 mk)

5. (a) The scheme below shows some of the reaction of solution **D**. Study it and answer the questions that follow



(i) Give a possible caution present in solution D (1 mk)

(ii) Write an ionic equation for the reaction in Step II (1 mk)

(iii) What observations would be made in Step V? Give a reason (2mks)

(iv) Explain why the total volume of hydrogen gas produced in step 1 was found to be very low although calcium and solution **D** were in excess. (2mks)

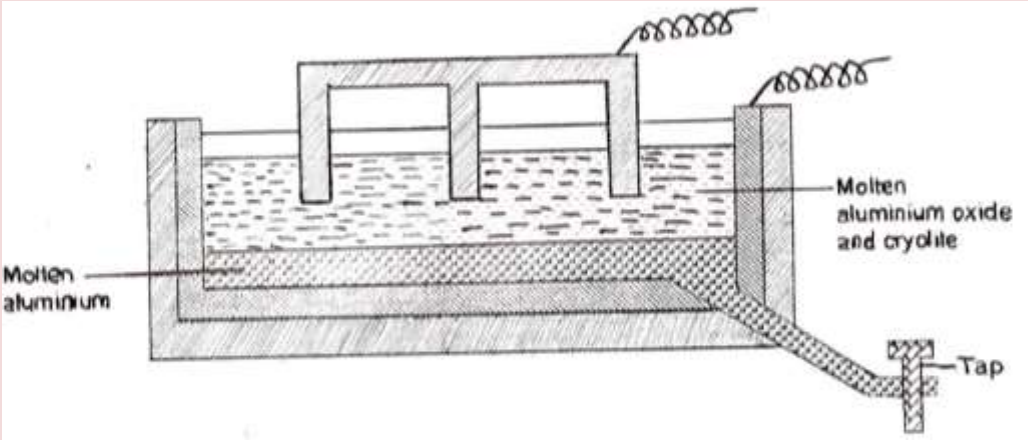
(v) State **one** use of substance **E**. (1 mk)

(b)Starting with solid sodium chloride, describe how a pure sample of lead (II) Chloride can be prepared in the laboratory (3mks)

(c) (i) State a property of anhydrous calcium chloride which makes it suitable for use as a drying agent for chlorine gas. (1 mk)

(ii) Name another substance that can be used to dry chlorine gas (1 mk)

6. The diagram below represents a set up of an electrolytic cell that can be used in the production of aluminium



(a) One the diagram, label the anode (1 mk)

(b) Write the equation for the reaction at the anode (1 mk)

(c) Give a reason why the electrolytic process is not carried out below 950^0C (1 mk)

(d) Give a reason why the production of aluminium is not carried out using reduction process (1 mk)

(e) Give **two** reasons why only the aluminium ions are discharged (2 mks)

(f) State two properties of duralumin that makes it suitable for use in aircraft industry (1 mk)

(g)Name two environmental effects caused by extraction of aluminium (2 mks)

7a) Dissolving of potassium nitrate in water is an endothermic process. Explain the effect of increase in temperature on the solubility of potassium nitrate (2 mks)

.

b) The table below shows the solubilities of potassium sulphate and potassium chlorate (V) at different temperatures.

Temperature (^{0}C)	0	20	40	60	80	100
Solubility of K_2S0_4 G/100 g water	8.0	10.0	14.0	17.5	20.0	22.0
Solubility of $KC10_3$ g/100g water	3.0	5.0	15.5	24.0	38.0	53.0

a) Draw the solubility curves for both salts on the same axis. (Temperature on the X-axis

(3 mks)

ii)A solution of potassium sulphate contains 20g of the salt dissolved in 100 g of water at 100^{0}C. This solution is allowed to cool to 25^{0}C

I) at what temperature will crystals first appears?

II) What mass of crystals will be present at 25^{0}C? (1mk)

iii) Which of the two salts is more soluble at 30^{0}C? (1mk)

iv) Determine the concentration of potassium sulphate in moles per litre when the solubility of the two salts are the same (K= 39.0, O=16.0 ; S=32.0) (3 mks)

v) 100 g of water at 100^{0}C contains 19g of potassium sulphate and 19 g of potassium chlorate (V). Describe how a solid sample of potassium sulphate at 60^{0}C can be obtained (2 mks)

CHEMISTRY PAPER 233/2

K.C.S.E 2015 QUESTIONS

1. (a) (i) Carbon (IV) oxide is present in soft drinks. State two roles of carbon (IV) oxide in soft drinks. (1 mk)

(ii) Explain the observation made when a bottle containing a soft drink is opened. (2 mks)

(iii) Carbon (IV) oxide dissolves slightly in water to give an acidic solution. Give the formula of the acid. (1 mk)

(b) Zinc oxide can be obtained by heating zinc nitrate. A student heated 5.76 g of zinc nitrate.

(i) Write an equation for the reaction that occurred. (1 mk)

(ii) Calculate the total volume of gases produced.
(Molar gas volume is 24 dm^3; Zn = 65.4; O = 16.0; N = 14.0). (4 mks)

(iii) Identify the element that is reduced when zinc nitrate is heated. Give a reason. (2 mks)

2. (a) Draw the structure of the folio wing compounds. (2 mks)
   (i) Butanoic acid;
   (ii) Pent-2-ene.

   (b) Explain why propan-1-ol is soluble in water while prop-1-ene is not.
   (Relative molecular mass of propan-1-ol is 60 while that of prop-1-ene is 42).
   (2 mks)

   (c) What would be observed if a few drops of acidified potassium manganate (VII)
   Were added to oil obtained from nut seeds? Explain. (2 mks)

   (d) State one method that can be used to convert liquid oil from nut seeds
   into solid. (1mk)

   (e) Describe how soap is manufactured from liquid oil from nut seeds

   (f) 0.44 g of an ester A reacts with 62.5 cm^3 of 0.08 M potassium hydroxide
   giving an alcohol Band substance C. Given that one mole of the ester reacts
   with one mole of the alkali, calculate the relative molecular mass of the ester.
   (2 mks)

3. (a) Name the method that can be used to obtain pure iron (III) chloride
   from a mixture of iron (III) chloride and sodium chloride. (1 mk)

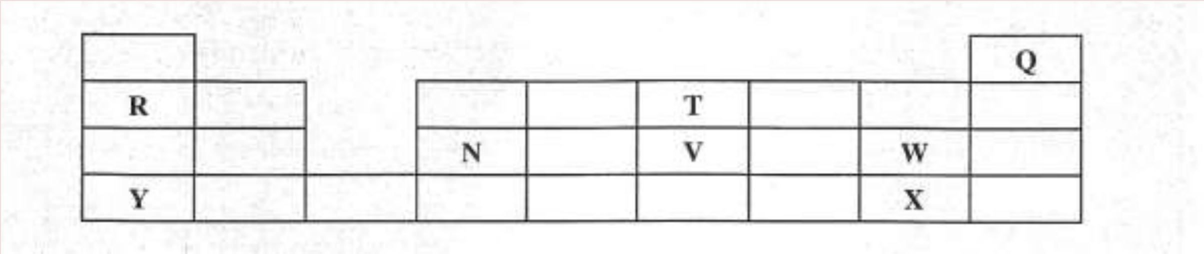
   (b) A student was provided with a mixture of sunflower flour, common salt and
   a red dye. The characteristics of the three substances in the mixture are given
   in the table below.

Substance	**Solubility in water**	**Solubility in ethanol**
Sunflower flour	Insoluble	Insoluble
Common salt	Soluble	Insoluble
Solid red dye	Soluble	Soluble

The student was provided with ethanol and any other materials needed.
Described how the student can separate the mixture into its three components

(3 mks)

c) The diagram below show part of a periodic table. The letters do no represent the actual symbols of elements. Use the diagram to answer the questions that follow.



i) Explain why the oxidizing power of W is more than that of X (2 mks)

ii) How do the melting points of R and T compare? Explain (2 mks)

iii) Sketch an element that could be used

i) In weather ballons (1 mk)

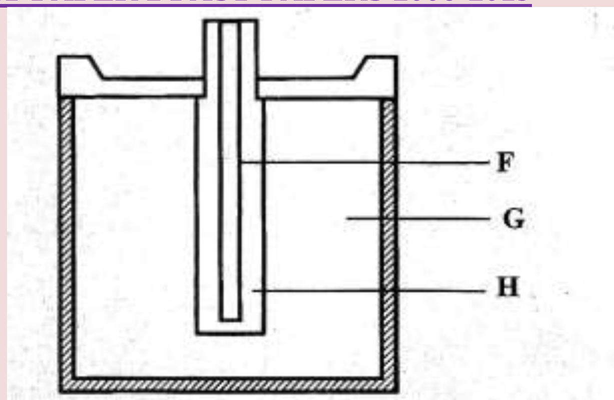
ii) For making a cooking pot (1 mk)

d i) Classify the substances water, iodine, diamond and candle wax into elements and compounds (2 mks)

Elements	Compounds

ii) Give one use of diamond (1 mk)

4a) The diagram below represents a dry cell. Use it to answer the quest ions that follows.



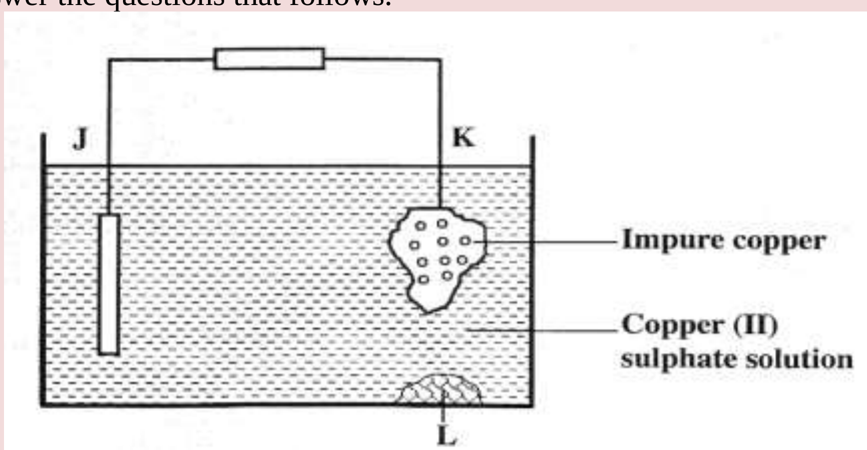
i) Which of the letters represent

i) Carbon electrode? (1 mk)

ii) The electrolyte? (1 mk)

ii) One of the substances used in a dry cell is manganese (IV) oxide. State two roles of manganese (IV) oxide in the dry cells (2 mks)

b) Below is simplified electrolytic cell used for purification of copper. Study it and answer the questions that follows.



i) Identify the cathode (1 mk)

ii) Write the equation for the reaction at the anode (1 mk)

iii) What name is given to L? (1 mk)

iv) A current of 0.6 A was passed Through the electrolyte for 2 hours.

Determine the amount of copper deposited

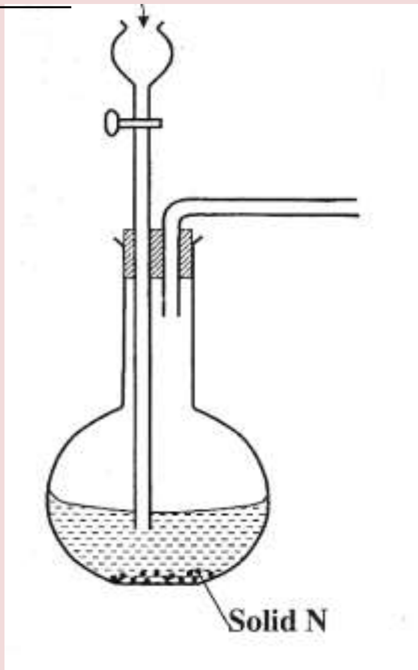
(Cu=63.5; 1 Faraday = 96,500 coulombs) (3 mks)

v) St ate two uses of copper metal (1 mk)

5. The set up below can be used to generate a gas without heating. This occurs when

 Substance M

substance Mr reacts with solid N.



a i) Complete the table below giving the names of substance M and solid N if the gasses generated are chlorine and sulphur (IV) oxide. (2 mks)

	Chlorine	Sulphur (IV) Oxide
Substance M		
Solid N		

(ii) Complete the diagram above to show how a dry sample of sulphur (IV) oxide can be collected

(b) Describe two chemical methods that can be used to test the presence of sulphur (IV) oxide. (3 mks)

(c) Other than the manufacture of sulphuric (VI) acid, state two uses of sulphur (IV) oxide. (2 mks)

6. (a) Other than concentration, state two factors that determine the rate of a reaction. (2 mks)

b) In an experiment to determine the rate of reaction, excess lambs of calcium carbonate were added to 2 M hydrochloric acid. The mass of calcium carbonate left was recorded after every 30 seconds. The results are shown in the table below

Time (seconds)	0	30	60	90	120	150	180	210
Mass of calcium carbonate left (g)	2.00	1.60	1.30	1.00	0.85	0.8	0.8	0.8

i) Write the equation for the reaction that took place

ii) On the grid provided, plot a graph of mass of calcium carbonate vertical axis Against time

(iii) Determine the rate of reaction at the 105th second. (3 mks)

(c) Why does the curve level off after some time? (1 mk)

(d) On the same grid, sketch a curve for the same reaction using 4 M hydrochloric acid and label the curve R. (2 mks)

7 (a) Naturally occurring magnesium consists of three isotopes. 78.6% ^{24}Mg; 10% ^{25}Mg and ^{26}Mg. Calculate to one decimal place, the relative atomic mass of magnesium. (2 mks)

(b) When magnesium burns in air, it forms a white solid and a grey-green solid. When a few drops of water are added to the mixture, a gas that turns red litmus paper blue is evolved.

Identify the

(i) white solid. (1 mk)

(ii) gas evolved and state its use.

(I) Name of gas (1 mk)

(II) Use of the gas.; (1 mk)

(c) Two different samples of water (I and II) were tested with soap solution. Sample II was further subjected to two other processes before adding soap. 20 cm^3 of each sample of water was shaken with soap solution in a boiling tube until a permanent lather was obtained. The results are shown in the table below

Water sample	Volume of soap solution needed (cm^3)	
	before boiling	after boiling
I	10	5
II	6	6
II after filtering	6	6
II after distilling	2	2

(i) Identify the water sample that had temporary hardness. Explain your answer. (2 mks)

(ii) Explain why the results for sample II are different after distilling but remain unchanged after filtering. (2 mks)

(iii) State two disadvantages of using both water samples for domestic purposes. (2 mks)

CHEMISTRY PAPER 233/2

K.C.S.E 2016 QUESTIONS

1. Use the information in the table below to answer the questions that follow, represent the actual symbols of the elements.

Element	**Atomic number**	**Melting point ^0C**
R	11	97.8

S	12	650.0
T	15	44.0
U	17	-102
V	18	-189
w	19	64.0

a) Give a reason why the melting pint of

i)S is higher than that of R (2 mks)

ii)V is lower than that of U. (2 mks)

b) How does the reactivity of W with chlorine compare with that of R with chlorine? (2 mks)

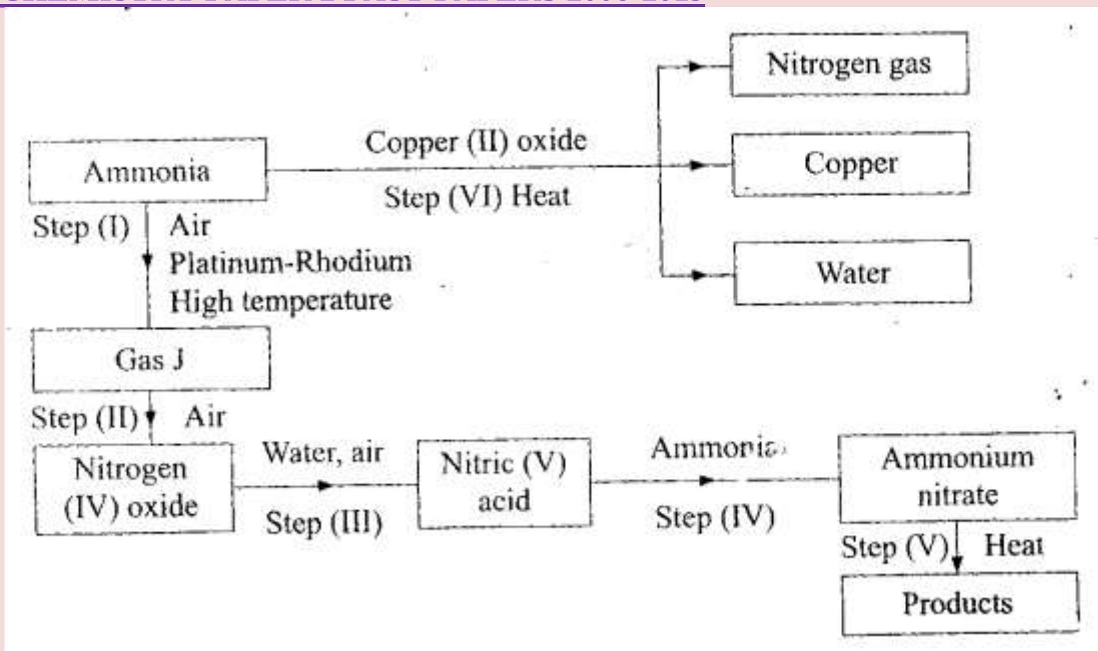
c) Write an equation for the reaction between T and excess oxygen (1 mk)

d) When 1.15 g of R was reacted with water 600cm^3 of gas was produced. Determine the relative atomic mass of R. (molar gas volume = 24000CM^3) (3 mks)

e) Give one use of element V (1 mk)

2. a) Describe the process by which nitrogen is obtained from air on a large scale (4mks)

b) Study the flow chart below and answer the questions that follow.



(i) Identify gas J. (1mk)

(ii) Using oxidation numbers show that ammonia is the reducing agent in step (VI) (2 mks)

(iii) Write the equation for the reaction that occurs in step (V). (1 mk)

(iv) Give two uses of ammonia nitrate. (2 mks)

(c) The table below shows the observation made when aqueous ammonia was added to cation of elements E, F and G until in excess

Cation of	Addition of a few drop s of aqueous ammonia	adiiton of excess aqueous ammonia
E	white precipitate	insoluble
F	no precipitate	no precipitate
G	white precipitate	dissolves

(i) Select the cation that is likely to be Zu^2+ (1 mk)

(ii) Given that the formula of the cation of element E is E^{2+} write the ionic equation for the reaction between E^{24} (aq) and aqueous ammonia. (1 mk)

3. (a) Methanol is manufactured from carbon (IV) oxide and hydrogen gas according to the equation,

$$CO_{2(g)} + 3H_{2(g)} \rightleftharpoons CH_3OH_3 + H_2O(g)$$

The reaction is carried out in the presence 'bf a chromium catalyst at 700K and 300k pa. Under these conditions, equilibrium is reached when 2% of the carbon (IV) oxide is converted to methanol.

(i) How does the rate of the forward reaction compare with that of the reverse reaction when 2% of the carbon (IV) oxide is converted to methanol? (1 mk)

(ii) Explain how each of the following would affect the yield of methanol:

I reduction in pressure (2 mks)

II using more efficient catalyst (2 mks)

(iii) If the reaction is carried out at 500k and 300k pa, the percentage of carbon (IV) oxide converted to methanol is higher than 2%.

I What is the sign of ΔH for the reaction? Give a Reason (2 mks)

II Explain why in practice the reaction is carried out at 700K but not at 500K (2 mks)

(b) Hydrogen peroxide decomposes according to the following equation:

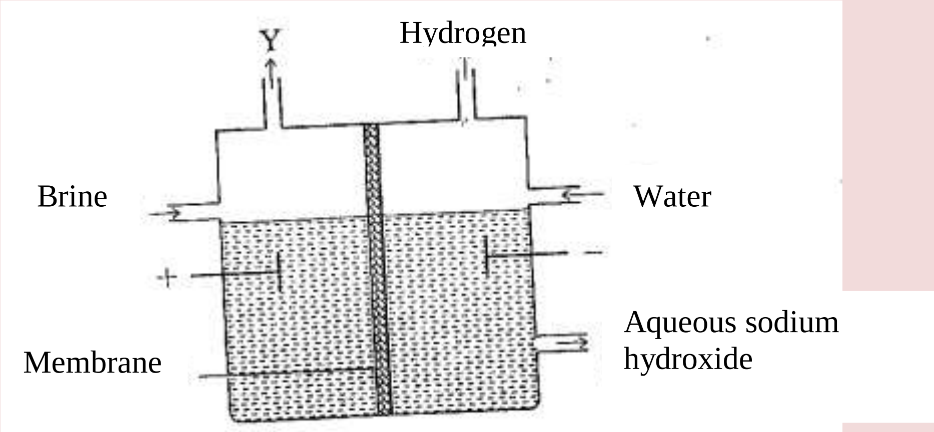
$$2H_20_2\ (aq) \longrightarrow 2H_2O_{(l)} + O_2(g)$$

In an experiment the rate of decomposition of hydrogen peroxide was found to be $6.0 \times l0^{-8}$ moldm $^{-3}$S^{-1}

(i) Calculate the number of moles per dm^3 of hydrogen peroxide that has decomposed within the first 2 minutes. (2 mks)

(ii) In another experiment, the rate of decomposition was found to be 1.8×10^{-7}moldm^{-3}S^{-1} . The difference in the two rates could have been caused by addition of a catalyst. State giving reason, one other factor that may have caused the difference in the two rates of decomposition. (2 mks)

4. a) The set up below can be used to produce sodium hydroxide by electrolyzing brine



i) Identify gas y (1 mk)

ii) Describe how aqueous sodium hydroxide is formed in the above set up (2 mks)

iii) One of the uses of sodium hydroxide is in the manufacture of soaps. State one other use of sodium hydroxide (1 mk)

b) Study the information given below and answer the questions that follow

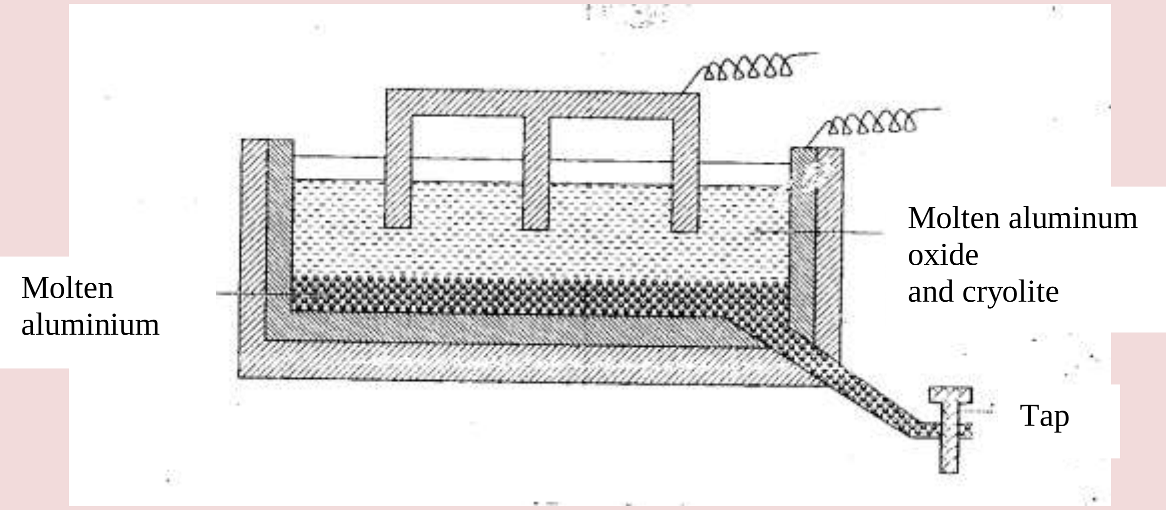
Half reactions	**Electrode potential $E^{\theta}V$**
$D^{2+}+_{(aq)} + e \longrightarrow D_{(s)}$	0.13
$E^{+}_{(aq)} + e \longrightarrow E_{(s)}$	+0.80
$F^{3+} + e \longrightarrow F^{2+}_{(aq)}$	+0.68
$G^{2+}_{(aq)} + 2e \longrightarrow G_{(s)}$	-2.87
$H^{2+} +(_{aq)} + 2e \quad H_{(s)}$	+0.34
$E^{+}_{(aq)} + e \longrightarrow E_{(s)}$	-2.71

(i) Construct an electrochemical cell that will produce the largest e.m.f.

(ii) Calculate the e.m.f. of the cell constructed in (i) above.

(iii) Why is it not advisable to store a solution containing E+ ions in a container in of H?

5. The diagram below represents a set up of an electrolytic cell that can be used in the production of aluminium.



(a) On the diagram, label the anode. (1 mk)

(b) Write the equation for the reaction at the anode. (1 mk).

(c) Give a reason why the electrolyte process is not carried out below 950^0C. (1 mk)

(d) Give a reason why the production of aluminium is not carried out using reduction process. (1 mk)

(e) Give two reasons why only the aluminium ions are discharged (2 mks)

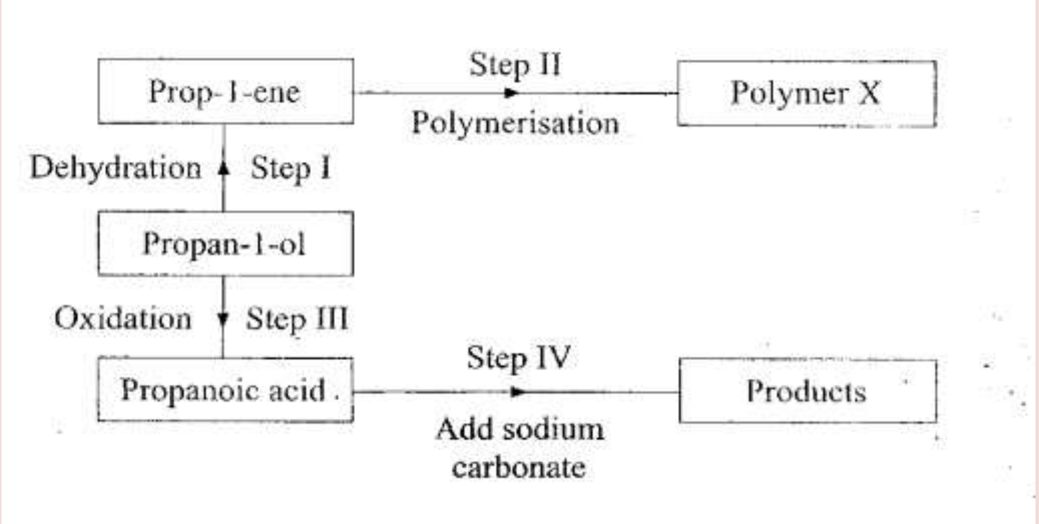
(f) State two properties of duralumin that makes it suitable for use in aircraft industry. (2 mks)

(g) Name two environmental effects caused by extraction of aluminum. (2 mks)

6. a)Draw the structural formula for all the isomers of $C_2H_3CL_3$ (2mks)

b) Describe two chemical tests that can be used to distinguish between ethene and ethane (4mks) c)The following scheme represents various reactions starting with propan -1-01.

Use it to answer the questions that follow



i)Name one substance that can be used in step 1 (1 mk)

ii) Give the general formula of X (1 mk)

(iii) Write the equation for the reaction in Step IV.

(iv) Calculate the mass of propan-1-ol which when burnt completely in air at room temperature and pressure would produce $18dm^3$ of gas. (C = 12.0, O = 16.0, H = 1.0; molar gas volume = $24dm^3$) (3 mks)

7. a) Write an equation to show the effects of heat on the nitrates of

i)Potassium (1mk)

ii) Silver (1 mk)

b) The table below gives information about elements A_1, A_2 , A_3 and A_4 .

Elements	Atomic Number	Atomic radius (nm)	Atomic radius (nm)
Al	3	0.134	0.074
A2	5	0.090	0.012
A3	13	0.143	0.050
A4	17	0.099	0,181

(i) In which period of the periodic table is element A ? Give a reason. (2 mks)

(ii) Explain why the atomic radius of:

I Al is greater than that of A_2 (2 mks)

II Al is smaller than its ionic radius. (2 mks)

(iii) Select the element which is in the same group as A3. (1 mk).

(iv) Using. Dots (.) and crosses (x) to outermost electrons, draw a diagram to show the bonding in the compound formed when Al reacts with A4. (2 mks)

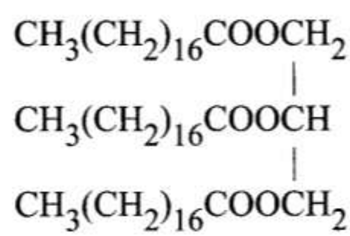
KCSE 2017 CHEMISTRY PAPER 2

1. (a) Name the homologous series represented by each of the following general formulae.

(i) CnH2n-2 (1 mark)

(ii) CnH2n (1 mark)

(b) Compound G is a triester.



Compound G

(i) Give the physical state of compound G at room temperature. (1 mark)

(ii) G is completely hydrolysed by heating with aqueous sodium hydroxide.

I Give the structural formula of the alcohol formed. (1 mark)

II Write a formula for the sodium salt formed. (I mark)

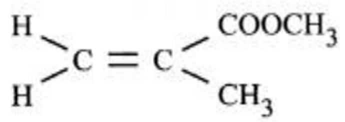
III State the use of the sodium salt. (1 mark)

(c) Ethyne is the first member of the alkyne family.

(i) Name two reagents that can be used in the laboratory to prepare the gas. (1 mark)

(ii) Write an equation for the reaction. (1 mark)

(d) Perspex is an addition synthetic polymer formed from the monomer,



(i) What is meant by addition polymerisation? (1 mark)
(ii) Draw three repeat units of perspex. (1 mark)

Give one use of perspex (1 mark)

(iv) State two environmental hazards associated with synthetic polymers. (1 mark)

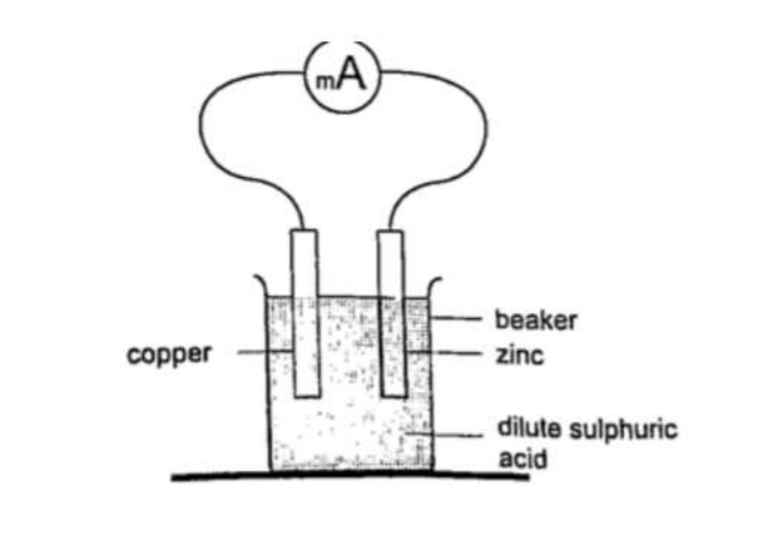
Table 1. Use it to answer the questions that follow.

Table 1

Substance	Conductivity in solid state	Conductivity in molten or aqueous state
F	Does not conduct	Conducts
G	Conducts	Conducts
H	Does not conduct	Does not conduct

2.(a) (i) Identify a substance that is a metal. Give a reason. (2 marks)
(ii) Substance F does not conduct electricity in solid state but conducts in molten or aqueous state. Explain. (2 marks)

(b) Copper(II) sulphate solution was electrolysed using the set up in Figure 1.



(i) State the observations made during electrolysis. (11/2 marks)
(ii) Write the equation for the reaction that occurs at the anode. (1 mark)

(iii) State the expected change in pH of the electrolyte after electrolysis. ('/2 mark)

(c) The experiment was repeated using copper electrodes instead of carbon electrodes.

Describe the observations made at each electrode. (1 mark)

(d) Electroplating is an important industrial process.

What is meant by electroplating. (1 mark)

(iii) During electroplating of an iron spoon, a current of 0.6 amperes was passed through aqueous silver nitrate solution for 11/2 hours. Calculate the mass of silver that was deposited on the spoon. (3 marks) (Ag = 108.0 ; I F = 96,500 C mo1-1)

3.(a) A student used Figure 2 to investigate the action of dilute sulphuric(VI) acid on some metals.Beaker I and II contained equal volumes of dilute sulphuric(VI) acid. To beaker I, a clean iron rod was dipped and to beaker II, a clean copper rod was dipped.

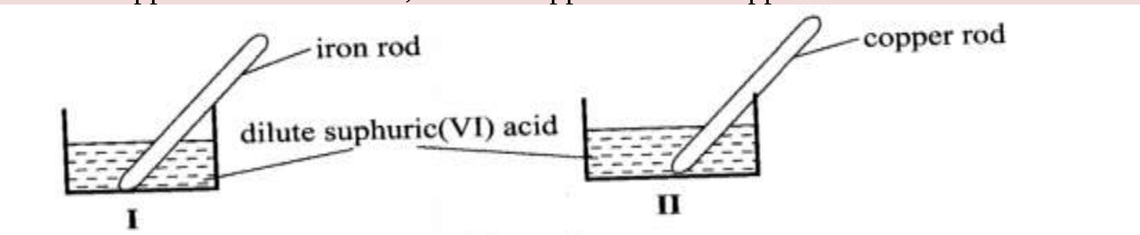


Figure 2

(i) Why was it necessary to clean the metal rods? (1 mark)
(ii) Describe the observations made in each beaker.

Beaker I: (1 mark)

Beaker II: (1 mark)

(iii) Explain the observations in (a) (ii). (2 marks)

(b) Figure 3 shows the apparatus used to burn hydrogen in air. Use it to answer the questions that follow.

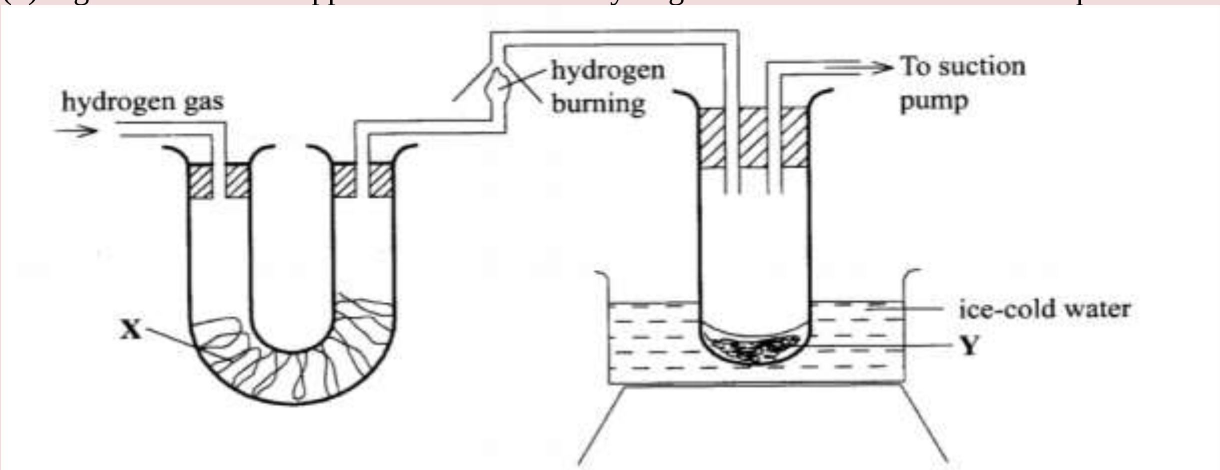


Figure 3

State the role of substance X.
ice-cold water Y (1 mark)

(ii) Give the name of the substance that could be used as X. (1 mark)

(iii) State the role of the suction pump. (1 mark)

(iv) Name the product Y formed. (1 mark)

(v) Give a simple physical test to prove the identity of Y. (1 mark)

vi) State the difference between 'dry' and 'anhydrous'. (2 marks)

4.W is a colourless aqueous solution with the following properties:

I It turns blue litmus paper red.

II On addition of cleaned magnesium ribbon, it gives off a gas that burns with a pop sound.

III On addition of powered sodium carbonate, it gives off a gas which forms a precipitate with calcium hydroxide solution.

IV When warmed with copper(II) oxide powder, a blue solution is obtained but no gas is given of

V On addition of aqueous barium chloride, a white precipitate is obtained.

(a) (i) State what properties (I) and (III) indicate about the nature of W. (I mark)

(ii) Give the identity of W. (l mark)

(iii) Name the colourless solution formed in (II) and (III). (2 marks)

(iv) Write an ionic equation for the reaction indicated in (V). (I mark)

(b) Element V conducts electricity and melts at 933K. When chlorine gas is passed over heated V, it forms a vapour that solidifies on cooling. The solid chloride dissolves in water to form an acidic solution. The chloride vapour has a relative molecular mass of 267 and contains 19.75% of V. At a higher temperature, it dissociates to a compound of relative molecular mass 133.5. When aqueous sodium hydroxide is added to the aqueous solution of the chloride, a white precipitate is formed which dissolves in excess alkali. (V = 27.0 ; CI = 35.5)

(i) Determine the:

I empirical formula (2 marks)

II molecular formula (2 marks)

(ii) Draw the structure of the chloride vapour and label the bonds. (1 mark)

(iii) Write an equation for the reaction that form a white precipitate with sodium hydroxide. (1 mark)

(a) When 0.048 g of magnesium was reacted with excess dilute hydrochloric acid at room temperature and pressure, 50 cm3 of hydrogen gas was collected. (Mg = 24.0; Molar gas volume = 24.0 dm3)

(i) Draw a diagram of the apparatus used to carry out the experiment described above. (3 marks)

(ii) Write the equation for the reaction. (1 mark)

(iii) Calculate the volume of hydrogen gas produced. (2 marks)

(iv) Calculate the volume of 0.1 M hydrochloric acid required to react with 0.048 g of magnesium. (3 marks) 6. The following steps were used to analyse a metal ore.

(i) An ore of a metal was roasted in a stream of oxygen. A gas with a pungent smell was formed which turned acidified potassium dichromate(VI) green.

(ii) The residue left after roasting was dissolved in hot dilute nitric(V) acid. Crystals were obtained from the solution.

(iii) Some crystals were dried and heated. A brown acidic gas and a colourless gas were evolved and a yellow solid remained. (iv) The solid was yellow when cold.

(v) The yellow solid was heated with powered charcoal. Shiny beads were formed.

Name the:

(a) gas formed when the ore was roasted in air. (1 mark)

(b) gases evolved when crystals in step (iii) were heated. (2 marks) (c) yellow solid formed in step (iii). (1 mark)

(d) shiny beads in step (iv). (I mark)

(e) The yellow solid from procedure (iii) was separated, dried, melted and the melt electrolysed using graphite electrodes.

1. Describe the observations made at each electrode. (2 marks)

II. Write the equation for the reaction that took place at the anode. (1 mark)

(f) Some crystals formed in step (ii) were dissolved in water, and a portion of it reacted with potassium iodide solution. A yellow precipitate was formed. Write an ionic equation for this reaction. (1 mark)

(g) To another portion of the solution from (f), sodium hydroxide solution was added drop by drop until there was no further change. Describe the observation made. (1 mark)

(h) To a further portion of the solution from (f), a piece of zinc foil was added.

I. Name the type of reaction taking place. (1 mark)

II. Write an ionic equation for the above reaction. (1 mark)

7. The decay rates of a sample of a radioisotope of bismuth at different time intervals is indicated in the following table.

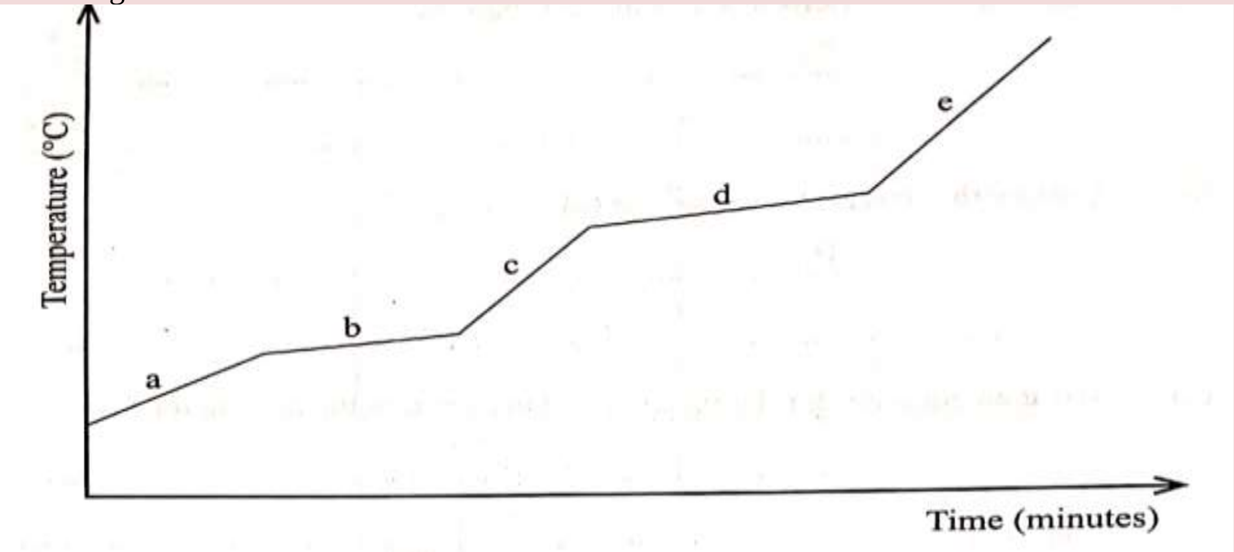


Figure 4

(ii) Determine the half-life of bismuth. (1 mark)
(iii) What would be the effect on the curve if half the amount of sample of bismuth were used. (1 mark)

14 (b) Radioactivity has several applications. State one application of radioactivity in: (i) Medicine (1 mark)

(ii) Agriculture (1 mark)

(iii) Tracers (I mark)

(iv) Nuclear power station (1 mark)

(c) State two dangers associated with radioactivity. (2 marks)

KCSE 2018 CHEMISTRY PAPER 2

1. The diagram in figure I shows some natural and industrial processes. Study it and answer the questions that follows

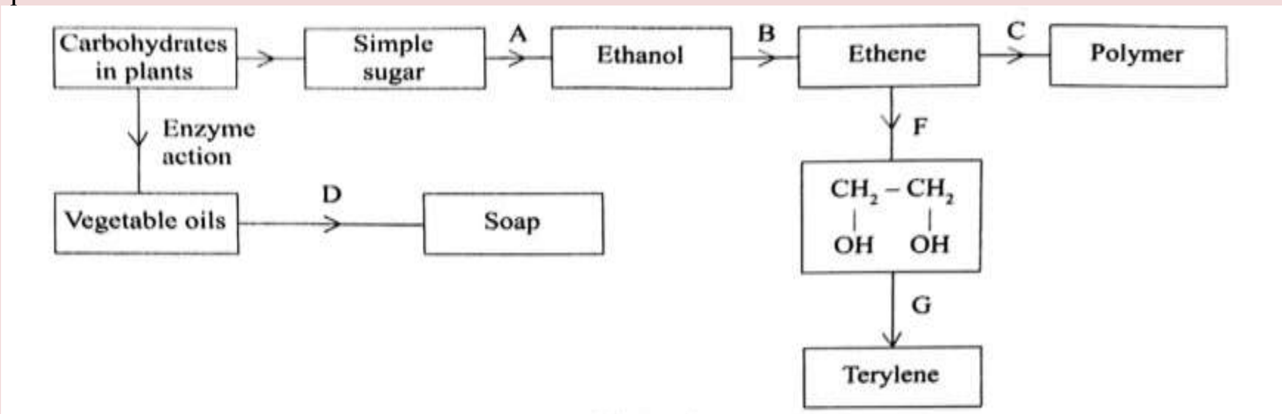


Figure 1

(a) Identify the processes labelled: (2 marks)
A..................

B..................

C..................

D..................

(b) State the reagents and conditions required for processes B and D.

(i) Process B:

Reagent................(1 mark)

Conditions .... (l mark)

(ii) Process D:

Reagent................(1 mark)

Conditions .... (l mark)

(iii) Describe how process D is carried out.(2 marks)

(iv) State two additives used to improve the quality of soap.(1 marks)

(c) State the reagents required in steps F and G.(1 marks)

(iii) Draw the structure of terylene.(1 marks)

(d) (i) Name the polymer formed in step C.(1 marks)

(ii) State one disadvantage of the polymer formed in (d) (i).(1 marks)

2. Figure 2 is a section of the periodic table. Study it and answer the questions that follow. The letters do not represent the actual symbols of elements

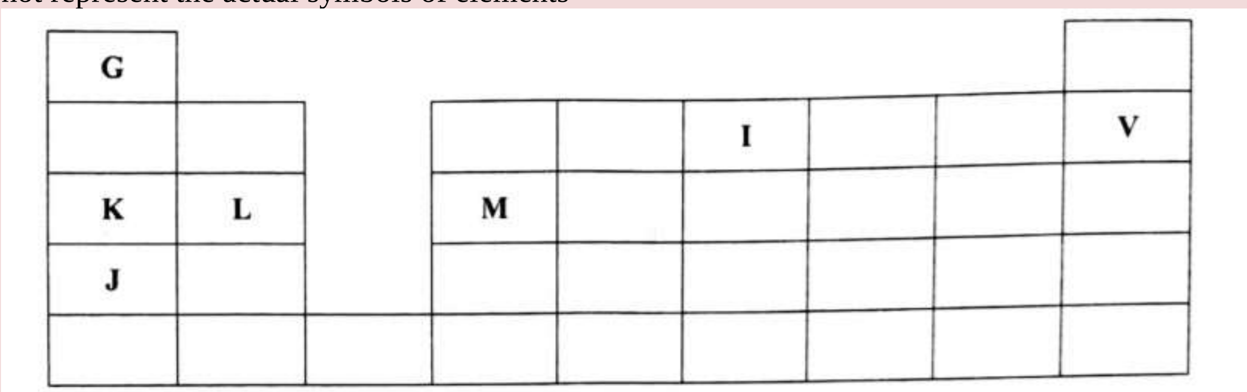


Figure 2

(a) (i) Select elements which belong to the same chemical family.(1 marks)
(ii) Write the formulae of ions for elements in the same period.(1 marks)

(b) The hrst ionisation energies of two elements K and M at random are 577 kJ/mol and 494 tel/mol .(1 marks)

(i) Write equations for the 1 st ionisation energies for elements K and M and indicate their energies.(1 marks)

(iii) Write the formula of the compound formed when L and I react.(1 marks)

(iv) Give one use of elemcnt V.(1 marks)

(c) (i) State anothcr group that G can be placcd in Figure 1. Explain. (2 marks)

(ii) How do the reactivity of elements J and K compare? Explain. (2 marks)

(d) (i) Elements L and M form chlorides. Complete the following table by writing the formulae of each chloride and state the nature of the solutions. (2 marks)

Element	Formula of chloride	Nature of chloride solution
L		
M		

(ii) The chloride of element M vapourises easily while its oxide has a high melting point. Explain. (2 marks)

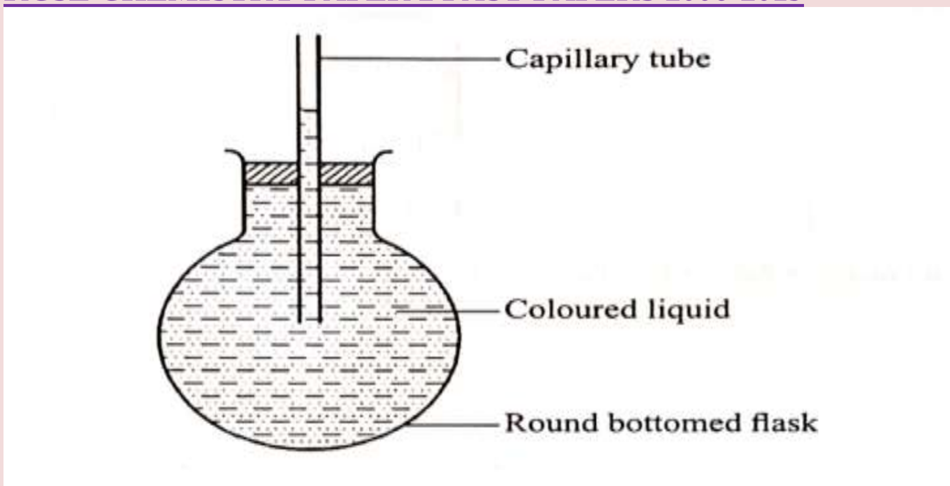
3. (a) Complete Table I by indicating the observations, type of permanent or temporary change and name of new compound formed.

Table 1

Experiment	Observations	Type of Change	Name of product
(i) Heat candle wax strongly in a test tube.			
(ii) Anhydrous copper(II) sulphate is left exposed overnight			
(iii) Iron wool is soaked in tap water for two days			

(6 marks)

(b) Use the set-up in Figure 3 to answer the questions that follow. The flask was covered with a cloth that had been soaked in ice-cold water.



(i) State the observation made on the coloured water. Explain.(2 marks)
(ii) Name the gas law illustrated in Figure 3.(1 marks)

(c) Use the standard electrode potentials in Table 2 to answer the questions that follow.

Table 2

Half-cell	E^{θ}/Volts
Z^{+}/Z	+0.80
V^{2+}/V	−0.40
W^{+}/W_2	0.00
Y^{2+}/Y	−2.87
U^{+}/U	+1.90

(i) Write the half-cell representation for the element whose electrode potential is for hydrogen. (1 mark)
(ii) Arrange the elements in order of reducing power, starting with the weakest reducing agent. (1 mark)

(iii) I Select two half cells which combine to give a cell with the least e.m.f. (1 mark)

II Calculate the e.m.f of the half cells identified in (iii) I. (1 mark)

4. An experiment was carried out to prepare crystals of magnesium sulphate.

Excess magnesi um powder was added to l00cm'of dilute sulphuric(VI) acid in a beaker and warmed until no further reaction took place.

The mixture was filtered and the filtrate evaporated to saturation, then lcft to cool for crystals to form.

(a) (i) Write an equation for the reaction.(1 mark)

(ii) Explain why eKcess magnesium powder was used.(1 mark)

(iii) State how completion of the reaction was determined.(1 mark)

(iv) What is meant by a saturated solution?(1 mark)

(v) Explain why the filtrate was not evaporated to dryness.(2 mark)

(b) When bleaching powder, CaOC1,, is treated with dilute nitric(V) acid, chlorine gas is released.

This reaction can be used to determine the chlorine content of various samples of bleaching powders and liquids.

(i) Write an equation for the reaction of nitric(V) acid with bleaching powder.(1 mark)

(ii) Calculate the volume of chlorine produced when 10 g of CaOC1, is treated with excess nitric(V) acid. (Ca = 40.0; O = 16.0; Cl = 35.5; 1 mole of gas occupies 22.4dm3 at s.t.p) (3 marks)

(c) Apart from use of chlorine gas in bleaches and water treatment, state two other uses of chlorine gas. (1 mark)

5. (a) The diagram in Figure 4 was used to prepare hydrogen chloride gas which was passed over heated iron powder.
iron powder.

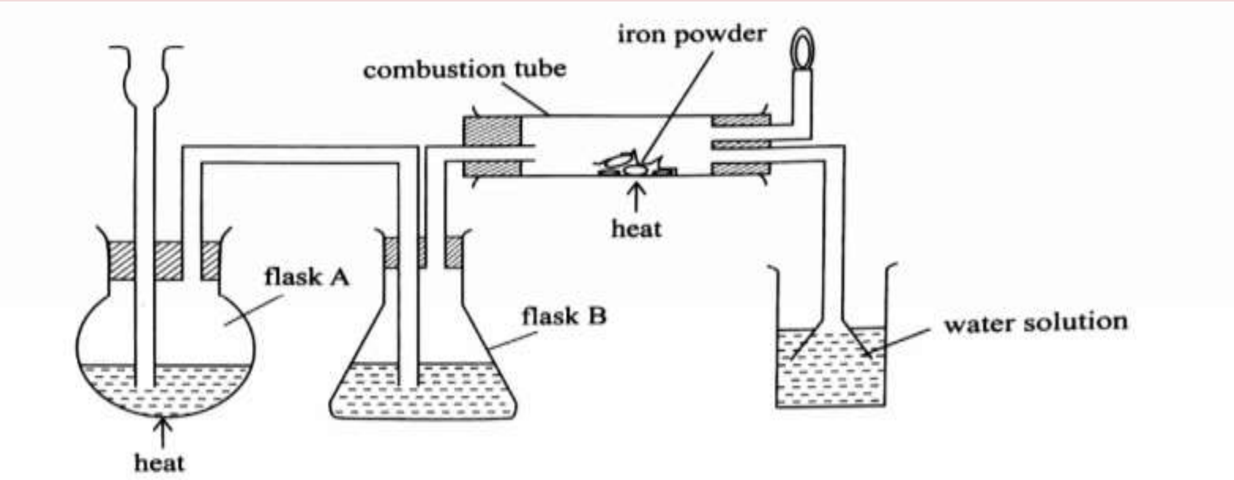


Figure 4

(i) Give a pair of reagents that will produce hydrogen chloride gas in flask A. (2 marks)
(ii) Name the substance in flask B.(1 marks)

(iii) State the observation made in the combustion tube.(1 marks)

(iv) Write an equation for the reaction in the combustion tube.(1 marks)

(v) Describe a chemical test for hydrogen chloride gas.(1 marks)

(b) (i) Identify the gas that bums at the jet.(1 marks)

(ii) Explain why the gas in (b) (i) is burned.(1 marks)

(c) Give reasons why excess hydrogen chloride gas is dissolved using the funnel arrangement. (2 marks)

(d) State what will be observed when the reaction in the combustion tube is complete.(1 marks)

(e) Another experiment was carried out where hydrogen chloride gas was bubbled through methylbeiizene and water in separate beakers.

The resulting solutions were tested with blue litmus papers and marble chips.

(i) Write the observations made in the following table.

Solution of hydrogen chloride gas in:	Blue litmus paper	Marble chips
Water		
Methylbenzene		

(2 marks)

(ii) Explain the observations in (e) (i).(2 marks)

6. (a) In Kenya, sodium carbonate is extracted from trona at Lake Magadi. (i) Give the formula of trona.

(ii) Name the process of extracting sodium carbonate from trona.(2 marks)

(b) The flow chart in Figure 5 summarises the steps involved in the production of sodium carbonate. Use it to answer the questions that follow.

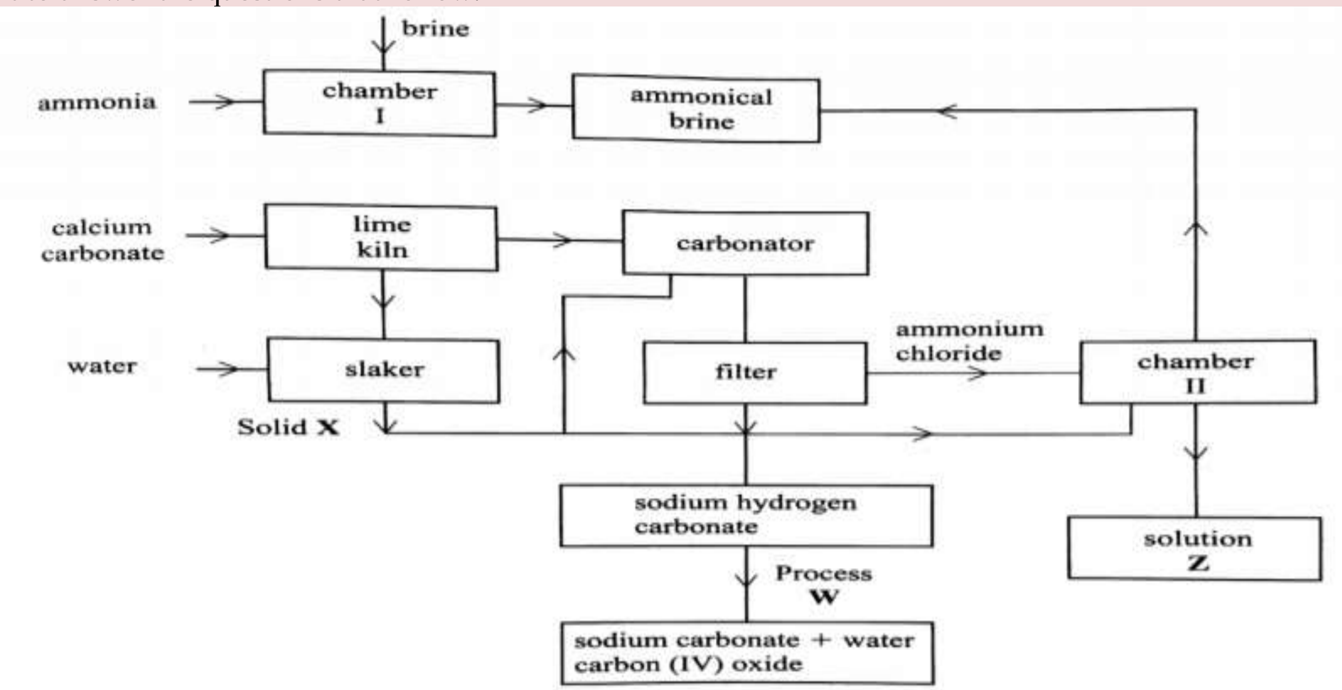


Figure 5

(i) Name the process illustrated in Figure 5.(1 mark)
(ii) Identify the starting raw materials required in the production of sodium carbonate. (2 marks)

(iii) Write equations for the two reactions that occur in the carbonator. (2 marks)

(iv) Name two substances that are recycled. (v) Identify:

Solid X;................(l mark)

Process W...............(l mark)

(vi) Write an equation for the reaction that produces solution Z.(l mark)

(vii) Apart from softening hard water, state two other uses of sodium carbonate. (2 marks)

KCSE 2019 CHEMISTRY PAPER 2

1. (a) Alkanes are said to be saturated hydrocarbons.

(i) What is meant by saturated hydrocarbons. (1 mark)

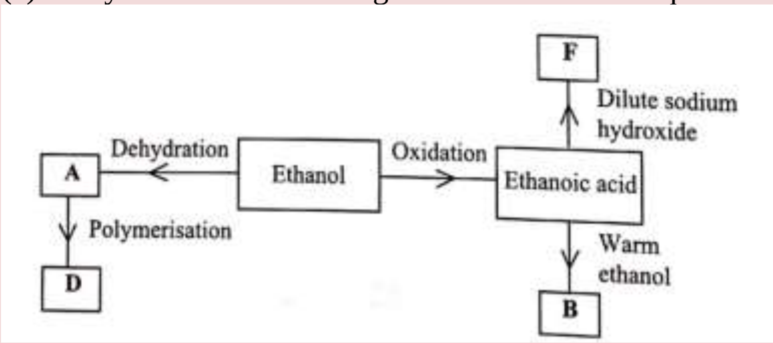
(ii) Draw the structure of the third member of the alkane homologous series and name it. (2 marks)

(b) When the alkane, hexane, is heated to high temperature, one of the products is ethene.

(i) Write the equation for the reaction. (1 marks)

(ii) Name the process described in (b). (1 marks)

(c) Study the flow chart in Figure 1 and answer the questions that follow.



(i) Identify A. (1 marks)
(ii) State one physical property of B. (1 marks)

(iii) Draw the structure of D. (1 marks)

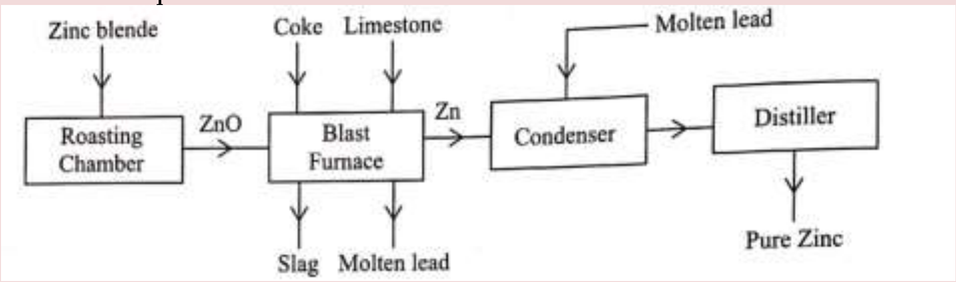
(iv) Give a reason why D pollutes the environment. (1 marks)

(v) Write an equation for the formation of F.(1 marks)

(d) Describe an experiment which can be used to distinguish butene from butanol. (2 marks)

2. (a) Zinc occurs mainly as zinc blende. Name one other ore from Which zinc can be extracted.(1 marks)

(b) The flow chart in Figure 2 shows the various stages in the extract oil of zinc metal. Study it and answer the questions that follow.



(i) Write an equation for the reaction which occurs in the roasting chamber. (1 mark)
(ii) Describe the process that takes place in the blast furnace. (3 mark)

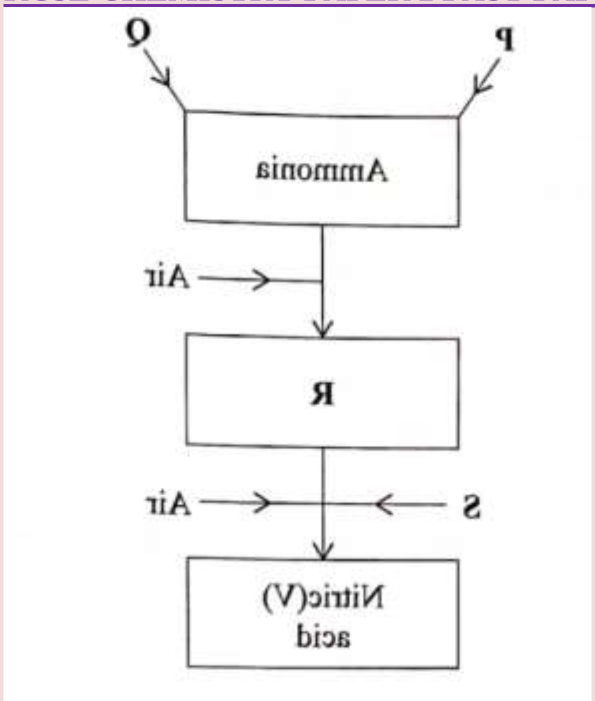
(iii) Explain why molten lead is added to the condenser. (1 mark)

(iV) State two uses of zinc.(1 mark)

(v) Give one reason why the extraction of zinc causes pollution to the environment. (1 mark)

(c) Explain the observations made when zinc metal is added to hot sodium hydroxide. (2 marks)

3. Figure 3 is a flow chart that shows the process that occurs in the manufacture of nitric(v)acid occurs in the manufacture of nitric(v)acid



(a) Name substance P, Q, R and S. P........................... (1 marks)
Q.............................. (1 marks)

R................................(1 marks)

S..................................(1 marks)

(b) To obtain substance R, ammonia is heated at 900 °C in the presence of air and a catalyst. The product is then cooled in air.

(i) Name the catalyst for the reaction. (1 marks)

(ii) Write the equations for the two reactions described in (b). (2 marks)

(iii) Other than nitric(V) acid, name another product that is formed. (1 mark)

(c) When ammonia is reacted with nitric(V) acid, it produces a nitrogenous fertiliser.

(i) Explain why fertilisers play a major role in food production. (2 marks)

(ii) State two problems associated with the use of nitrogenous fertilisers. (2 marks)

4. (a) Explain the following observations:

(i) The colour of aqueous copper(ii) sulphate fades when a piece of magnesium metal is dropped into the solution. (2 marks)

(ii) A piece of iron bar is coated with a brown substance when left in the open on a rainy day. (2 marks)

(b) A sample of water is suspected to contain aluminium ions (AI3+) .

Describe a laboratory experiment that can be carried out to show that AI3+ ions are present in the water sample. (3 marks)

(c) In an experiment to determine the number of moles of water of crystallisation of a hydrated compound Na2SO4.XHsO,5g of the compound were heated strongly to a constant mass.

(i) Explain how a constant mass was obtained. (2 marks)

(ii) During the experiment, the mass of the residue was found to be 2.205 g.

Determine the number of moles of water of crystallisation in the compound. (Na = 23.0 ; O = 16.0 ; S = 32.0 ; H = 1.0) (3 marks)

5. (a) What is meant by a molar heat of neutralisation? (1 marks)

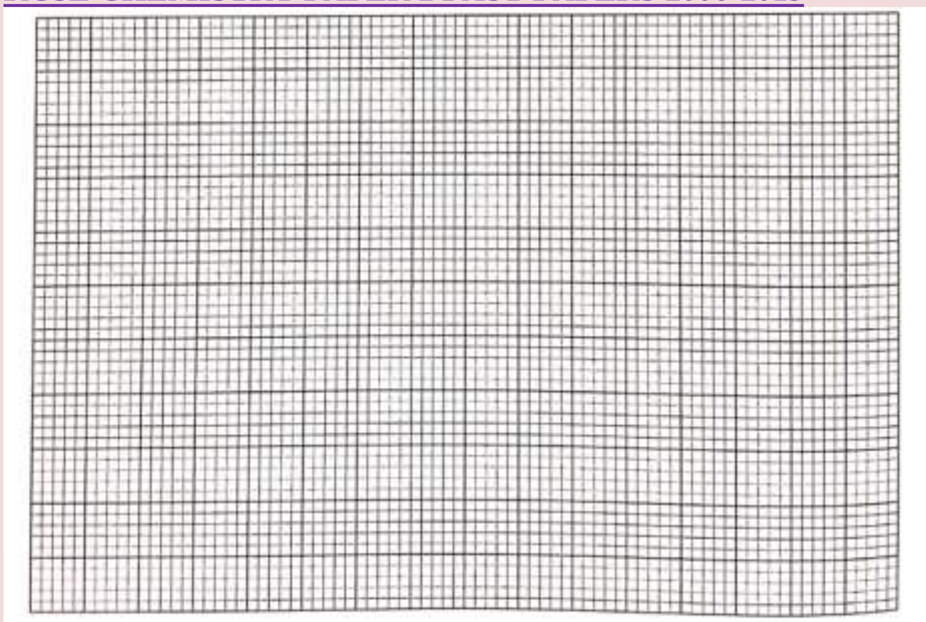
(b) In an experiment to determine the molar heat of neutralisation, 50 cm3 of IM hydrochloric acid was neutralised by adding 10 cm3 Onions Of dilute sodium hydroxide.

During the experiment, the data in Table 1 was obtained.

Table 1

Volume of Sodium hydroxide (cm^3)	0	10	20	30	40	50	60
Temperature of mixture (°C)	25.0	27.0	29.0	31.0	31.0	30.0	29.0

(i) Write the equation for the reaction in this experiment. (1 mark)
(ii) On the grid provided, plot a graph of temperature (Y-axis) against volume of sodium hydroxide (X-axis) added. (3 marks)



(iii) Determine from the graph the:
I. volume of sodium hydroxide which completely neutralises 50 cm3 of lM hydrochloric acid. (1 mark)

11. change in temperature, AT, when complete neutralisation occurred. (1 mark)

(IV) Calculate:

I. The heat change, dH when complete neutralisation occurred. (Specific heat capacity = 4.2 Jg-1 K-1 density of solution 1.0 gcm-3)(2 marks)

(v) How would the value of molar heat differ if 50 cm3 of lM ethanoic acid was used instead of IM hydrochloric acid? Give a reason. (2 marks)

6. (a) What is meant by standard electrode potential of an element (1 mark)

(b) Use the standard electrode potentials given below to answer the questions that follow.

Reactions	E^{θ} (V)
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$	+1.49
$M^{3+}(aq) + e^- \rightarrow M^{2+}(aq)$	+0.77
$N^{2+}(aq) + 2e^- \rightarrow N(s)$	+0.34
$P^{2+}(aq) + 2e^- \rightarrow P(s)$	−0.23
$Q_2(g) + 2e^- \rightarrow 2Q^-(g)$	+2.87
$R_2(g) + 2e^- \rightarrow 2R^-(g)$	+1.36

(i) State whether acidified MnO-4 can oxidise M2+. Give a reason. (2 marks)

(ii) Select two half-cells which when combined will give the highest e.m.f. (1 mark)

(iii) Write the cell representation for the cell formed in b (ii). (1 mark)

(iv) Calculate the Eθ value for the cell formed in b (iii). (1 mark)

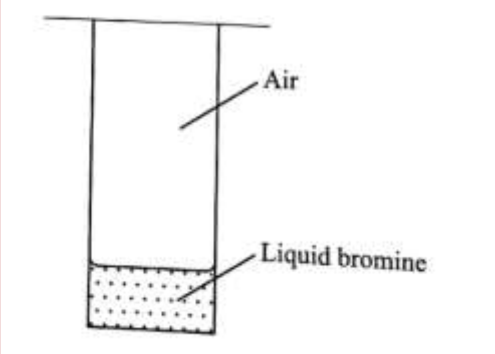
(c) A mass of 1.24g of a divalent metal was deposited when a current of 6A was passed through a solution of a metal sulphate for 12 minutes.Determine the relative atomic mass of the metal(Faraday = 96,500 C mol -1 (3 mark)

(d) State two application s of electrolysis. (I mark)

7. (a) What is meant by rate of reaction (1 mark)

(b) In the space provided, Sketch the diagram of a set-up that Can be used to determine the rate of reaction between manganese(IV) oxide and hydrogen peroxide. (3 marks)

(c) A student placed a small amount of liquid bromine at the bottom of a sealed gas jar of air as shown in Figure 4.



(i) Describe what will be observed: (1 mark)
I. After two minutes..............................

II. After 30 minutes ....

(ii) Use the Kinetic theory to explain the observations: (2 marks)

I. After two minutes..............................

II. After 30 minutes ....

(d) Some plants have seeds that contain vegetable oil.

(i) Describe how the oil can be obtained from the seeds. (3 marks)

(ii) Explain how it could be confirmed that the liquid obtained from the seeds is oil. (1 mark)